

RMS880/950 Documentation

Offline PDF edition of the nicFW880/950 documentation wiki.

Source wiki: <https://github.com/nicsure/RMS880/wiki>

Contents

Welcome to the nicFW880/950 Documentation Wiki

RMS Overview

RMS Serial

Channels Tab Overview

Groups Tab Overview

User Keys Tab Overview

Band Plan Tab Overview

Scanning Tab Overview

DTMF Tab Overview

GPS / APRS Tab Overview

Settings Tab Overview

Codeplug Tab Overview

Power Tab Overview

Calibration Tab

Remote Tab Overview

Fonts Tab Overview

UI Tab Overview

Startup Tab Overview

Console Tab Overview

Firmware Tab Overview

macOS and Linux

Radio Mode Overview

Tuner Mode

Spectrum Scope Mode

GPS Mode

KISS Mode Overview

Channel Menu Overview

CMS (Channel Management System) Menu

Squelch Overview

Scanning Menu Overview

VFO Menu Overview

Calibration Menu Overview

Display Menu Overview

User Keys Menu Overview

Advanced Menu Overview

DTMF Menu

Time Menu Overview

GPS Menu

APRS Menu

Band Plan Menu

Storage Deferral Overview

Channel Groups Overview

Scanning Overview

GPS Waypoints
T9 Text Entry Overview
NOAA Tone Detection
CAT Commands
Serial Protocol Overview
Remote Protocol Overview
Storage Flash Memory Map
All values are little endian

Welcome to the nicFW880/950 Documentation Wiki

RMS

- General Operation
- Serial Ports and BAUD rate
- Channels
- Groups
- User Keys
- Band Plan
- Scanning
- Presets
- Exclusions
- Smart List
- DTMF
- GPS/APRS
- Waypoints
- Beacons
- Settings
- Codeplug
- Power
- Calibration
- Remote
- Fonts
- UI
- Startup
- Console
- Firmware
- MacOS & Linux

Radio

Operating Modes

- Radio
- Tuner
- Scope
- GPS
- Cross Band Repeater
- KISS

Main Menu

- Channel
- CMS
- Squelch
- Scanning

- VFO
- Calibration
- Display
- User Keys
- Advanced
- DTMF
- Time
- GPS
- APRS
- Band Plan
- Shut Down

Operation

- Channel Groups
- Scanning

VFO Scan

Channel & Group Scan

Frequency Counter

- GPS Waypoints
- APRS
- Storage Deferral
- T9 Text Entry
- NOAA
- Factory Reset

When the radio is turned off, press and hold:

(RT-880) EMG (Top Button) (RT-950) SK3 (Bottom Side Button) while turning the radio on.

Wait until the display tells you the wipe process is complete.

Turn the radio off and then back on.

Technical Information

- CAT Commands
- Serial Protocol
- Remote Protocol
- Flash Storage Layout
- Data Structures

RMS Overview

The RMS (Radio Management System) is the companion desktop application for the radio firmware. It provides a powerful and user-friendly way to configure, manage, and back up virtually every aspect of the radio without needing to navigate menus on the device itself.

Using the RMS, you can configure channels, groups, scanning behavior, band plans, APRS/GPS features, power output, UI skins, key mappings, and more. Changes are made on the computer and then written to the radio over a serial connection, making complex configuration tasks faster, clearer, and far less error-prone than performing them directly on the radio.

RMS versions are tightly coupled to firmware versions. Unless explicitly stated otherwise in a release post, users should always download and use the matching RMS version that corresponds to the firmware installed on the radio. Mismatched versions may result in missing features or incorrect behaviour.

Download links for the RMS are provided alongside firmware releases and are available for Microsoft Windows, Linux, and macOS.

RMS Operation Overview

The RMS is organized into a series of tabs, with each tab responsible for configuring a specific aspect of radio operation. You can work with each tab independently, adjusting only the settings you are interested in.

There is also a Codeplug tab, which allows you to combine many common configurations into a single file. This is useful for creating backups, sharing configurations, or moving a complete setup between radios. See: RMS -> Codeplug

An important concept is that the RMS does not immediately apply changes to the radio. Instead, it maintains its own internal copy of the radio's storage, think of this as a working draft.

As you change settings in the RMS, you are modifying this draft copy, not the radio itself. To apply those changes to the radio, you must explicitly write them.

Configuration Files

All configuration files loaded and saved by the RMS use the .csv (Comma-Separated Values) format. This is a plain-text format that can be opened and viewed with common tools such as spreadsheet applications or text editors.

Using a text-based format was a deliberate design choice. Files with clearly labeled, human-readable fields are far easier to inspect, edit, back up, and troubleshoot than opaque binary files.

For power users, this makes advanced manipulation and automation straightforward. For everyday users, these files behave just like any other configuration file; load them, save them, and move them around as needed, without requiring any special knowledge of how they work internally.

RMS Tabs - Common Controls

Most RMS tabs include the same four buttons. Understanding what these do will make using the RMS much easier.

Load

Loads a previously saved .csv file into the RMS. This updates the RMS's working copy and the on-screen settings, but does not change anything on the radio.

Save

Creates a .csv file from the current RMS working copy. This is useful for backups or sharing settings. This action does not read anything from the radio.

Read

Reads the relevant settings from the radio into the RMS. This synchronizes the RMS working copy with what is currently stored on the radio.

Write

Writes the RMS working copy back to the radio. This is the step that actually applies your changes to the device.

Common Tab Tasks

Here's some examples of common tasks you might perform with the RMS tabs.

Backup of tab configuration

- Switch to the tab responsible for the settings you wish to backup.
- Click Read to synchronize the RMS to the radio.
- Click Save to create and save a .csv file to your computer.

Restoring tab configuration from backup

- Switch to the tab responsible for the settings you wish to restore.
- Click Load and select the .csv file created previously.
- Click Write to apply the backup to the radio.

RMS Serial Ports & Baud Rate

Communication between the RMS and the radio occurs over a serial programming cable.

- (RT-880 Only) Although the RT-880 includes a USB-C port, a standard USB-C to USB-A cable only supports battery charging and cannot be used for data communication.

A USB cable specifically designed for the RT-880 or RT-950 is required. These are available from Raddtel and various online marketplaces. Search for: RT-880 Programming Cable. RT-950 Pro Programming Cable. Note: The RT-950 Pro generally comes supplied with a programming cable in the box.

It is recommended to use cables with quality chipsets, such as CH340. Avoid cheaper cables using Prolific chipsets, as they are often unreliable.

Port / Baud Controls

The bottom-right corner of the RMS application window contains the serial configuration controls. These are labeled Port/Baud and consist of two drop-down selectors.

text

Port Dropdown

Displays all serial ports currently detected by your computer.

Select the port associated with your RT-880 programming cable.

If multiple ports are listed, an easy way to identify the correct one is:

1. Open the port dropdown
2. Unplug the programming cable
3. Note which port disappears
4. Plug the cable back in and select that port

Baud Dropdown

Allows selection of the communication speed used between RMS and the radio.

Available baud rates range from:

- 38,400 up to 518,400

The maximum reliable speed depends on:

- Cable quality
- Radio tolerances
- Computer hardware and drivers

Some experimentation may be required to determine the fastest stable setting.

Firmware Flashing Note

The configurable baud rate does not apply to firmware flashing. Firmware flashing always uses a fixed baud rate of 115,200.

macOS Note

Warning: Important for macOS users

On macOS versions Tahoe 26.3 or earlier, 38,400 baud is the only working speed.

Faster speeds are working with Tahoe 26.4 or later. You may need to experiment to find the fastest reliable speed for your system.

Channels Tab Overview

The Channels tab provides advanced tools for configuring and organizing radio channels and channel groups. It is the RMS equivalent of the radio's Main Menu -> CMS section, but with significantly expanded capabilities.

Channel List and Selection

On the left side of the tab is a scrollable list displaying all 999 channel slots.

- Click a channel to select it
- Hold CTRL while clicking to select multiple individual channels
- Hold SHIFT while clicking to select a continuous range of channels

When one channel is selected, all of its settings are displayed in the controls on the right.

When multiple channels are selected, only settings that are common to all selected channels are shown. Any setting that differs between channels will appear blank. Editing a setting will set all selected channels at once.

Channel 999

This can be a special channel reserved for APRS use. This is the channel to configure with your APRS operating frequency. See: Main Menu -> APRS -> Enabled

CHIRP Import and Export

The RMS supports importing from and exporting to the CHIRP channel CSV format.

CHIRP Import (CHIRP -> RMS)

You may load a CHIRP-format channel .csv file into the RMS. When imported:

- The RMS converts the file for use with nicFW
- All channels defined in the file from 1 to 999 are loaded
- Channels outside this range are ignored

CHIRP Export (RMS -> CHIRP)

Exporting converts all currently defined channels in the RMS into a CHIRP-format .csv file. The exported file may then be loaded into CHIRP or any other application that supports the CHIRP CSV format.

This makes it easy to move channel data between RMS, CHIRP, and other compatible tools.

Channel List Right-Click Menu

The channel list provides a context (right-click) menu for managing channels efficiently. Note that actions like Insert or Delete that shift blocks of channels up or down will not affect channel 999, this channel is always left alone.

Up

Moves the selected channel(s) up by one slot.

Down

Moves the selected channel(s) down by one slot.

Cut

Erases the selected channel(s) and places them into the clipboard.

Copy

Copies the selected channel(s) into the clipboard without removing them.

Paste

Overwrites channel slots starting at the selected position with the contents of the clipboard.

Insert

Shifts all channels from the selected slot down by one position, erasing channel 998 if necessary, and leaves the selected slot empty.

Delete

Erases the selected channel and shifts all subsequent channels up by one position, leaving channel 998 empty.

Sort Ascending / Descending

Reorders a contiguous block of selected channels based on RX frequency:

- Ascending: Low to high frequency
- Descending: High to low frequency

Channel Settings

For detailed explanations of these settings, see: Main Menu -> Channel

Active

Must be enabled for a channel to exist.

- Unticking this option effectively erases the channel

Name

A descriptive name for the channel.

- Maximum length: 30 characters
- The number of visible characters depends on the radio's UI configuration

When multiple channels are selected, the % symbol may be used for automatic numbering:

- Example: PMR %
- > Generates PMR 1 through PMR 5 for five selected channels
- Example: GMRS %15
- > Starts numbering at 15 instead of 1

See:

- RMS -> UI
- Main Menu -> CMS -> Channel Name

RX / TX

Defines the receive and transmit frequencies (in MHz).

- Entering an RX frequency automatically updates TX to match, preserving any existing offset

When multiple channels are selected, a frequency step may be specified:

- Example: 144.00000+12.5

-> RX frequencies increment by 12.5 kHz per channel

For split TX frequencies:

- Absolute TX: enter full frequency (e.g. 146.520)
- Relative offset:
 - +1.5 -> TX is 1.5 MHz above RX
 - -2.75 -> TX is 2.75 MHz below RX

Clarifier / BFO

Dual-purpose setting:

- Radio channels: Fine frequency adjustment in kHz for off-frequency stations
- Tuner channels: BFO offset used for SSB and CW reception

RX Tone / TX Tone

Configures receive and transmit squelch subtones. If RX Tone is set to the custom value 105.0 this enables NOAA Tone Detection for the channel.

- Supports CTCSS and DCS
- Standard and non-standard tones may be entered manually
- Right-click the field to select from a predefined list

Groups

Assigns group membership for the channel.

- Up to four groups per channel
- Groups are labeled A-Z

Modulation

Dual-purpose setting:

- Radio channels: FM, AM, DSB, or Automatic
- Automatic uses the band plan default
- Tuner channels: WFM, AM, USB, LSB, CW

See:

- RMS -> Band Plan

Bandwidth

Dual-purpose setting:

- Radio channels: Wide, Narrow, or Automatic
- Automatic uses the band plan default
- Tuner channels: 1-6 kHz

See:

- RMS -> Band Plan

TX Power

Sets transmit output power for the channel.

Options:

- N/T (No Transmit)

- Zero
- Very Low
- Low
- Medium
- High
- Very High

Zero is a special diagnostic mode that performs a transmit cycle with little to no RF output. This is useful for testing transmit functions like PTT-ID or Roger Beep without actually broadcasting with any significant output.

PTT ID

Controls whether the PTT-ID DTMF sequence (DTMF preset slot #99) is transmitted.

Options:

- Off
- Start of TX
- End of TX
- Both

Busy Lock

When enabled, prevents transmission if the channel is currently busy.

Reversed / AM Filter

Dual-purpose setting:

- Radio channels: Sets or clears the RX/TX reverse flag
- Note: This does not swap RX and TX values in the RMS; it only toggles the flag
- Tuner channels: Enables or disables the Si4732 power line filter

Scrambler

Selects the split-and-invert frequency used by the built-in voice scrambler.

- Applied to transmitted audio
- Decoded on receive when enabled

Note: Scrambling may be illegal on amateur or public-access bands. Use responsibly.

Groups Tab Overview

The Groups tab allows you to assign names to the 26 channel groups, labeled A through Z. This tab is the RMS equivalent of the radio-side Group Name menu function. Each group is displayed alongside a text input field. Enter the desired name into the field to assign or change the name for that group. Group names may be up to 12 characters in length. See: Main Menu -> CMS -> Group Name

Codeplug Inclusion

Group name settings are included when creating or loading a Codeplug, allowing them to be backed up, shared, or transferred together with other radio configuration data.

See: RMS -> Codeplug

User Keys Tab Overview

The User Keys tab allows you to assign functions to key presses used in Radio Mode. This tab is the RMS equivalent of the radio-side User Keys menu. Each assignable key is listed along with a drop-down control. Use the drop-down to select the function you want to assign to that key.

See:

- Main Menu -> User Keys

Fixed Keys

The following keys have fixed functionality and cannot be reassigned:

RT-880(G)

- SP-UP
- SP-DOWN
- SP-0 to SP-9
- SP-RED
- SP-GREEN
- SP-
- SP-#

RT-950 Pro

- SP-UP
 - SP-DOWN
 - SP-LEFT
 - SP-RIGHT
 - SP-0 to SP-9
 - SP-GREEN
 - SP-ABC
 - SP-VM
-

Band Plan Tab Overview

The Band Plan defines a set of preconfigured frequency bands that apply default settings and behavior to any frequency that falls within their defined ranges.

This tab is the RMS equivalent of the radio's Band Plan menu.

Band Slots

On the left side of the tab is a list of 20 band plan slots, numbered 0 through 19.

Selecting a slot from the list populates the controls on the right side of the tab. These controls allow you to edit the parameters that define how the radio behaves when operating within that frequency range.

Deleting a Band Slot

To delete a band slot, set its Start frequency to 0. This effectively disables the slot.

Additional Information

For a detailed explanation of each Band Plan setting and how it affects radio behaviour, see:

Main Menu -> Band Plan

Scanning Tab Overview

The Scanning tab provides tools for configuring and managing scan-related behaviour used by the radio. It allows you to define reusable scan ranges, manage ignored (excluded) frequencies, and inspect the frequency memory used by the Smart Scan system.

This tab is divided into three sub-tabs:

- Presets - pre-configured frequency scan ranges
- Exclusions - the scan ignore list
- Smart List - frequencies remembered by Smart Scan

For detailed information on how scanning behaves on the radio itself, see:

- Radio -> Operation -> Scanning

Presets Tab

The Presets tab manages the 99 available scan presets.

A list of all presets appears on the left side of the tab. Selecting a preset populates the controls on the right with that preset's configuration.

Preset Controls

For more details on these scan settings see: Main Menu -> Scanning

Active

Enables or disables the selected scan preset.

- Unticking this option effectively deletes the preset

Name

A descriptive name for the scan preset.

- Maximum length: 14 characters

Start

The starting frequency of the scan range, in MHz.

End

The ending frequency of the scan range, in MHz.

Step

Defines the spacing between scanned frequencies, in kHz.

Mode

Selects the modulation mode used during the scan.

Bandwidth

Selects the bandwidth setting used during the scan.

Ultra Scan

Controls the sensitivity of the Ultra Scan acceleration system.

Monitor Time

Defines how long (in seconds) a detected frequency will be monitored before the scan continues.

- A value of 0 causes the scan to remain on the frequency until the signal drops

Hold Time

Defines how long (in seconds) the scan will pause after a signal is lost before resuming. This can be useful for catching replies during conversations.

Exclusions Tab

The Exclusions tab displays the list of up to 50 ignored frequencies or channels currently stored in the radio.

Ignored frequencies or channels are added automatically when the user presses # on the radio during a scan. Once added, these are skipped during future scans, which is useful for avoiding nuisance signals.

Selecting a slot allows the frequency or channel to be edited, created, or cleared using the controls on the right.

- Setting a slot's frequency & channel to 0 disables it and frees the slot for reuse
- Main Menu -> Scanning -> Save Ignores must be set to On for this tab to function.

Smart List Tab

When Smart Scan is enabled, the radio remembers frequencies that were detected as busy during scans and revisits them more frequently. This can significantly increase the number of scan hits across large frequency ranges.

The Smart List tab allows these remembered frequencies to be:

- Read from the radio
- Inspected and exported
- Copied and pasted into the Channels tab

This provides a convenient way to turn frequently active scan hits into permanent channels.

DTMF Tab Overview

The DTMF tab allows you to pre-configure up to 99 DTMF sequences for later use. DTMF usage, transmission behaviour, and related radio-side settings are documented in detail here:

Main Menu -> DTMF

Preset Slots

All 99 DTMF preset slots are displayed in a list on the left side of the tab.

To create, edit, or delete a preset:

1. Select a slot from the list on the left by clicking on it. It will highlight to indicate it is selected.
2. The slot's current information (if any) will appear in the controls on the right
3. Use the controls to configure the selected slot

As changes are made, the slot list updates in real time to reflect the current configuration.

Slot #99 is reserved and is used to define the DTMF PTT-ID to be used when PTT-ID is active.

- See: Main Menu -> Channel -> PTT ID

Slot Controls

Active

Enables or disables the selected preset slot.

- A slot must be active before it can be edited
- Deactivating a slot effectively deletes its contents

Name

Sets a descriptive name for the preset.

- Maximum length: 14 characters

Digits

Defines the DTMF digit sequence for the preset.

- Valid characters: 0-9, *, #, A-D
- Maximum length: 14 digits

DTMF presets configured here are available for use throughout the radio wherever DTMF functions are supported.

GPS / APRS Tab Overview

The GPS / APRS tab is divided into two sub-tabs:

- Waypoints - for managing pre-programmed GPS locations
- Beacons - for viewing received APRS packets

Together, these tabs provide tools for navigation, location spoofing, and APRS monitoring.

Waypoints Tab

Waypoints are pre-programmed GPS coordinates that can be used as:

- Navigation targets
- Spoofed GPS locations
- Saved points of interest while navigating

The radio supports 99 waypoint slots, displayed in a scrollable list on the left side of the tab.

- Click a slot to select it
- When selected, the controls on the right populate with that waypoint's data

Reserved Slot

Slot #99 is reserved as the default spoofed location. When the radio starts with GPS functionality disabled, waypoint #99 is automatically used as the radio's GPS position.

Waypoint Tab Controls

Active

Enables or disables the waypoint.

- Unticking this box effectively deletes the waypoint

Latitude

The waypoint's latitude in degrees.

Longitude

The waypoint's longitude in degrees.

Name

Assigns a descriptive name to the waypoint.

- Maximum length: 14 characters

MH Locator

The Maidenhead Locator for the waypoint.

- Editing this field with a valid locator automatically updates the Latitude and Longitude
- Editing Latitude or Longitude updates the Maidenhead Locator
- Maidenhead locators are less precise than direct coordinate entry

Waypoint List Right-Click Menu

Copy / Cut / Paste

Standard clipboard operations for copying, moving, or deleting waypoints.

Browse

Opens a web browser centered on the selected waypoint's location using OpenStreetMap.

Additional Resources

A video guide covering the Waypoints tab is available here:

- Waypoints Guide

For more information, see:

- Main Menu -> GPS

Beacons Tab

The Beacons tab displays APRS beacons received by the radio.

- Beacons are listed on the left side of the tab
- The most recent beacon appears in slot #1
- Older beacons follow in chronological order

Beacons can be selected and copied into the Waypoints tab, allowing received APRS locations to be saved as navigation targets.

Beacons are not editable, you may only read them from the radio. You can however erase them all from the radio by pressing the Wipe button.

For more information on APRS beacons, see:

- APRS -> Beacons

Settings Tab Overview

All storage based settings that can be edited using the radio's Main Menu may also be configured in this tab. Every menu category and all the settings in that category are available.

For a full description of each setting and their corresponding values See Radio -> Main Menu

Codeplug Tab Overview

The Codeplug tab allows multiple RMS configuration sections to be combined into a single file. The following tabs are included:

- Channels
- Groups
- User Keys
- Band Plan
- Scanning
- DTMF
- GPS / APRS
- Settings

This makes it possible to perform a complete backup or restore of most radio configuration data in one step, which is significantly faster and more convenient than managing each tab individually.

Excluded Tabs

The following tabs are not included in the Codeplug:

- Power
- Calibration
- Fonts
- UI

These sections are excluded because they either:

- Contain binary data that is not compatible with the .csv format
- Store device-specific calibration values
- Control visual or aesthetic preferences rather than operational configuration

This separation helps ensure Codeplugs remain portable, safe to share, and broadly compatible across devices.

Power Tab Overview

The Power tab provides advanced control over the radio's RF output characteristics. It allows experienced users to fine-tune the five transmit power levels across different frequency ranges.

The available power levels are:

- Very Low
- Low
- Medium
- High
- Very High

See Main Menu -> Channel -> TX Power

Frequency Slots

Frequency coverage is divided into 5 MHz slots, spanning from 0 MHz to 1300 MHz. These slots are listed on the left side of the interface.

You do not need to configure every slot. When transmitting, the radio automatically selects the closest configured slot relative to the tuned frequency. In practice, assigning a single slot near the centre of the band you wish to adjust is usually sufficient.

Power Parameters

Once a frequency slot is selected, the following parameters can be adjusted independently for each of the five power levels.

Bias

Controls the power amplifier (PA) bias voltage.

- Range:
- 0 = 0.0 V to
- 255 = 3.2 V

Gain 1 & Gain 2

Adjusts the RF gain tuning stages used by the power amplifier.

- Range: 0 to 7

Technical Info

The Bias and Gain values map directly to internal BK4819 registers. Refer to the BK4819 Application Notes, specifically Register 0x36.

Usage Notes

Correctly tuning these values requires experimentation and measurement using appropriate test equipment, such as a dummy load and power meter.

For most users, it is strongly recommended to retain the default values provided after a fresh firmware flash or factory reset. Improper configuration can result in poor performance, distortion, or hardware stress.

Calibration Tab Overview

The Calibration tab contains radio-specific settings used to fine-tune behaviour that can vary slightly between individual devices due to manufacturing tolerances.

These settings are intended for advanced users and should be adjusted with care.

Battery and Oscillator Options

Battery calibration and oscillator calibration options mirror those available on the radio itself and are documented in the Radio Mode manual.

See:

- Main Menu -> Calibration

Band (Hz Offset)

Band offset calibration is only configurable through the RMS.

These settings apply a fine-grained frequency correction in Hertz on a per-band basis. Each band spans a 50 MHz range, starting at 0 MHz and extending up to 1300 MHz.

The defined offset is applied to all frequencies that fall within the selected band.

Example

Setting a value of -300 in the band labelled <150 MHz applies a -300 Hz frequency correction to any tuned frequency between 145 MHz and 150 MHz.

This feature allows precise correction for band-specific frequency drift that cannot be accurately resolved using a single global oscillator adjustment.

RSSI Floors

Defines the signal floor for specific frequency ranges. These floor values are used as the reference point for calculating S-meter readings.

Because receiver sensitivity can vary slightly across different frequency bands, nicFW allows separate floor values to be configured per range. Setting these correctly ensures that S-meter readings are consistent and meaningful across the spectrum.

The values entered here correspond to the internal register units of the BK4819.

Converting to dBm

To convert the configured value to dBm:

1. Divide the value by 2
2. Subtract 160

Example

For a setting of 100:

- $100 \div 2 = 50$
- $50 - 160 = -110$ dBm

So a value of 100 corresponds to -110 dBm.

S-Meter Scaling

Each S-Unit on the signal meter represents 6 dBm above the configured noise floor defined in this section.

Accurate floor calibration ensures that:

- S-meter readings align with expected signal strengths

- S-levels remain consistent across different frequency bands

The radio will display the current received signal level in the same units while the Squelch menu is open. This can be useful for quick reference.

Remote Tab Overview

The Remote tab allows the radio to be fully controlled from within the RMS. The left side of the tab displays a live remote view of the radio's screen, while the right side presents a button panel that mirrors the radio's keypad and function keys.

Starting a Remote Session

To start a remote control session:

1. Toggle the switch located beneath the S2 button to the On position
2. When the connection is successful, the status bar will display "Remote Started"

Starting the remote session does not automatically refresh the screen. If the display does not appear immediately, open the radio's main menu and then close it to force a screen refresh.

Remote Operation

Once connected:

- Clicking buttons in the RMS behaves exactly as if the corresponding physical key were pressed on the radio
- All changes to the radio's display are reflected in the remote viewer
- The radio's physical keys remain fully usable during remote operation

The mouse wheel is mapped to the UP and DOWN keys:

- Scrolling up acts as UP
- Scrolling down acts as DOWN

Keyboard Shortcuts

In addition to mouse interaction, the following keyboard shortcuts are available:

Keyboard Key	Radio Button
Cursor Up Arrow / Page Up	UP
Cursor Down Arrow / Page Down	DOWN
Enter / Return	GREEN
Escape / Backspace	RED
0-9 (including numeric keypad)	0-9
Spacebar	PTT
Tab (and # on UK keyboards)	#
Period / Multiply / Asterisk	
F1	S1
F2	S2
F3	EMG

Note Regarding PTT Usage with the Remote

The radio emits a significant amount of RF energy during transmission. This energy can interfere with nearby electronic devices in a manner similar to an EMP, potentially causing instability.

Possible effects include:

- Computers or laptops freezing, crashing, or restarting
- USB serial interfaces malfunctioning or disconnecting
- Keyboards and mice becoming unresponsive
- Other unpredictable behaviour in nearby electronics

If issues occur when engaging PTT via the remote interface, RF interference is the most likely cause.

To reduce these effects:

- Increase the distance between the radio and your computer or laptop
- Use ferrite clamps on the USB programming cable
- Use an external cable fed antenna rather than one mounted directly on the radio.
- Use a lower power setting

Fonts Tab Overview

The Fonts tab allows visual customization of the radio by assigning different textual fonts. These fonts are bitmapped, monospaced fonts encoded in column-major format. In this format, bit data is stored column by column, from left to right and top to bottom.

Supported Font Sizes

The following font sizes can be customized:

- 8?8 (ASCII)
- 8?16 (ASCII)
- 16?16 (ASCII)
- 16?24 (ASCII)
- 24?24 (ASCII)
- 24?32 (ASCII)
- 16?16 Symbols (Custom Icons)

Interface Layout

- The left side of the tab displays a live preview of all fonts currently loaded into the RMS.
- The right side contains the font control panel.

Each font entry is labelled with its pixel dimensions and includes two associated buttons.

Font Controls

Load

The Load button allows you to browse for a font file compatible with the selected font size.

- The RMS will reject files that do not match the expected file size
- Once loaded, the preview pane updates immediately
- This action loads the font into the RMS only and does not modify the radio

Export

The Export button saves the currently loaded font from the RMS to the local machine. This can be used to back up fonts or share them with others.

Radio Synchronization

Below the individual font controls are two general buttons used to synchronize fonts with the radio.

Read

Reads all font data previously written to the radio into the RMS.

- The preview pane updates to reflect the fonts currently stored on the radio
- Note: Do not use this function if no custom fonts have been written to the radio

The radio?s default fonts are stored in ROM and are not readable. The RMS starts with these default fonts loaded by default, so reading is unnecessary in that case.

Write

Writes all currently selected fonts in the RMS to the radio. This overrides the default fonts

or replaces any previously written custom fonts on the device.

Codeplug Note

Fonts are stored as binary data and are therefore not compatible with the combined Codeplug format.

UI Tab Overview

The UI tab is the most complex section of the RMS. It allows you to completely redesign the appearance of the radio's main display by repositioning visual elements, selecting fonts, and defining colors.

The process of designing or editing the radio's interface is referred to as skinning. UI designs, along with the .csv files saved and loaded by this tab, are commonly called skins. Several example skins are included in the extras.zip file available from the RMS release page. Skinning is not an easy task. It can be very time consuming, fiddly and tedious, so be warned before diving into this rabbit hole.

Color Palette

On the left side of the tab is a scrollable list of predefined color functions. These colors do not simply represent static colors, they represent logical functions that change based on the radio's state.

Each color entry is labeled with the function it represents.

Example

The colors RX Idle, RX Open, and TX define the state colors for the active VFO:

- RX Idle - Squelch closed
- RX Open - Squelch open (receiving)
- TX - Transmitting

Any UI element whose foreground or background color is set to VFO State will automatically use the appropriate color based on the current radio state.

To change a color, click on the example color box to open a color editor dialog.

Preview

To the right of the color palette is the preview display. This is a live mock-up of the radio's screen.

As you change colors, positions, fonts, or modes, the preview updates immediately to show how the display will look under different operating conditions.

Element List

To the right of the preview is a scrollable list containing all visual elements that make up the display.

- Click an element to select it
- Once selected, all of its properties are loaded into the control panel on the right
- Only one element may be edited at a time

Control Panel

The far-right section of the tab is the control panel. This is where the selected element's properties are edited and where various preview simulation options are available.

Control Panel Items

Name Label

At the top of the control panel is the name of the currently selected element.

Cursor Arrows

Convenience buttons used to move the selected element left, right, up, or down. These are often easier to use than manually editing coordinates.

X

The horizontal position of the element.

Note:

For VFO child elements, this value is relative to the parent VFO Box element's X position.

Y

The vertical position of the element.

Note:

For VFO child elements, this value is relative to the parent VFO Box element's Y position.

Width & Height

These values are rarely required for text-based elements, as their size is usually fixed by the font.

Some elements, such as VFO Box and User Rects do require explicit width and height values.

- Setting Width to 0 disables the element entirely and prevents it from being drawn

FG Color & BG Color (Default)

Defines the default foreground and background colors for the element.

- Setting a color to Alpha makes it transparent.
- Note: Transparency is a bit of a misnomer, it merely means to use the same color as the the VFO's background color.
- These colors are used when no color function is defined or when the radio state does not match a defined function (for example, inactive VFOs)
- To change, click the color example box to open a color editing dialog.

FG Color Function & BG Color Function

Causes the element's background or foreground color to change based on a particular radio state. Use the pulldown list to select the particular radio state to associate with the element. The colors for each individual state can be edited using the color palette on the left hand side of the tab.

Justification

For text-based elements, defines whether the text is left-justified or right-justified.

Font

Selects which font is used to render the selected text-based element.

Signal / Power Level

Controls the signal level or TX power level displayed by the preview's signal meter.

Noise / Mod Level

Controls the noise bar level shown in the preview.

Battery Level

Sets the battery level displayed in the preview.

The battery element respects the battery-level color functions when rendering.

VFO Status

Changes the preview to simulate different radio states:

- Squelch Closed - Idle state
- Squelch Open - Receiving a signal
- TX - Transmitting
- Scanning - Active scan
- Scan Monitor - Monitoring a detected scan hit

VFO Mode

Changes how the preview represents different VFO operating modes:

- VFO - Frequency mode
- Channel - Channel mode
- Group - Group mode

Channel Name Length

Defines the maximum number of characters that the Channel Name element can display without overlapping other elements or running off the right of the screen.

This is particularly important when using right justification. It is also crucial for the correct operation of scrolling channel names.

Menu Open

Simulates the appearance of the main menu in the preview.

Menu Selected

When the menu is displayed, simulates how the UI looks while editing a menu option.

Reset Button

Discards all changes and restores the UI to the default layout.

Important Notes

- Do not read UI data from the radio unless a skin has already been written to it
- Reading UI data from a radio without a skin will return invalid data
- There is no need to read the default UI from the radio, as the RMS starts with the same default skin preloaded

Startup Tab Overview

The Startup tab allows you to customize the image displayed when the radio powers on.

The boot image may be in any common image format, but it must be exactly 240 x 320 pixels to be accepted.

Setting a Boot Image

To configure a boot image:

- Click the Load Image button and select a suitable image from your computer
- A preview of the image will appear on the left of the window.
- Set Logo Display Seconds to define how long the image should remain on screen
- Click Write to Radio to apply the image to the device

The new boot image will be shown the next time the radio is powered on.

Console Tab Overview

The Console tab provides low-level serial communication with the radio. Two data modes are available: Text and Hex.

This tab is primarily intended for developers and advanced users who wish to test CAT commands and programming packets. It may also be used as a debug console at the request of the developer to help diagnose difficult or intermittent issues.

Ensure the correct Serial Port is selected before starting a console session.

While a console session is active you may not switch tabs until you end the session.

Text Console

To start a Text Console session, click the Text Console button. The border of the text console window will turn orange, and the text input box below will become active. You may then enter commands into the input box. The console window will display any text returned by the radio.

- See: Technical Info -> CAT Commands

The developer may occasionally request that you activate this console and report any output for debugging purposes.

To terminate the session, click the Text Console button again. The orange border will disappear, and the text input box will be deactivated.

Hex Console

To start a Hex Console session, click the Hex Console button. The border of the hex console window will turn orange, and the input box below will become active. You may then enter raw hexadecimal data to communicate directly with the radio.

Hexadecimal data must be entered as pairs of hexadecimal digits. Example: 94 A7 2B AF

The console window will display data received from the radio in the same hexadecimal format.

- See Technical Info -> Serial Protocol

The developer may occasionally request that you activate this console and report any output for debugging purposes.

To terminate the session, click the Hex Console button again. The orange border will disappear, and the input box will be deactivated.

Firmware Tab Overview

The Firmware tab provides tools for updating the radio's firmware, as well as backing up and restoring the radio's internal storage memory.

A video guide on performing firmware flashing and storage backup operations can be viewed here:
Video Guide

Firmware Flashing

Select Radio Model

- Under Select Radio Model, choose the radio model you are using:
- Raddtel RT-880
- iRadio UV-98

Note:

For Raddtel radios with a serial number below 300, you must select iRadio UV-98.

Select Firmware File

Under Select Firmware File, click Browse and select the firmware binary (.bin) file you wish to flash.

Preparation

Before flashing:

- Ensure the radio is connected to the computer using an approved RT-880 or UV-98 programming cable
- Select the correct COM port for your radio in the bottom-right corner of the RMS window

Note: The baud rate setting is not used for firmware flashing

Enter Flash Mode

1. Turn off the radio
2. Power it on while holding the PTT button
3. The radio's screen should remain black and the green LED should illuminate

Flashing

Click Flash Firmware and wait for the process to complete.

Important:

When flashing nicFW, the progress bar will reach 100% and then timeout.

This is normal behaviour. Simply power the radio off and back on to boot into the new firmware.

When flashing stock firmware, the radio will automatically reboot when flashing completes.

Storage Backup

Taking a Backup

It is strongly recommended to take a full backup of the radio's storage chip immediately after installing nicFW, before pressing any buttons or interacting with the radio in any way. This backup allows you to fully restore the radio to its original state later.

1. After nicFW boots for the first time, do not press any buttons
2. Select a reasonably fast baud rate
 - This is a long operation; values around 345,600 are typically suitable
 - Reliable baud rates may vary between radios and may require experimentation
3. Click Read Flash Image
4. Wait for the block counter in the status bar to count down to zero
5. Click Save Flash Image and store the backup file in a safe location

Important:

After completing a flash backup, close the RMS completely and restart it before continuing.

Restoring a Backup

To restore a previously saved flash image:

1. Ensure the radio is powered on and running nicFW
2. Click Load Flash Image and select your saved backup file
3. Click Write Flash Image and wait for the block counter to reach zero
4. Power off the radio

You may now flash the stock firmware back onto the radio (see Firmware Flashing above). Once complete, the radio will be fully restored to its factory state.

If you did not take a backup image, you can download ones that I have taken from new radios here:

- Flash Backup Images

macOS and Linux

RMS builds for macOS and Linux are available and linked from the nicFW release posts.

Because the RMS is developed on Windows, packaging non-Windows releases with the correct file permissions and operating-system-specific flags is more complex. As a result, Linux and macOS releases are provided primarily as .zip archives containing the required binaries and libraries. After extracting these files, users must manually set the appropriate execution permissions.

This process is usually familiar to Linux users but can be less intuitive for macOS users. For this reason, macOS releases are also provided in a standard App Bundle format alongside the .zip files. However, Mac users must still deal with quarantining and third party app permissions.

Regarding macOS specifically

At present, the ARM64 (Apple Silicon) builds of the RMS for macOS are not functional. Users should instead use the x64 version, which works correctly under Rosetta. ARM64 builds continue to be compiled and released in the hope that the issue can be identified, either by an experienced user or through future fixes in Avalonia, the application framework used by the RMS.

For reasons that are still under investigation, macOS Tahoe 26.3 or earlier currently operate only at 38,400 baud, however macOS Tahoe 26.4 or later fix this issue and faster speeds are available.

Radio Mode Overview

Radio mode is the default state of the radio when first powered on, the classic ?Walkie-Talkie? experience.

The radio mode operates utilising a Radio System on Chip, the Beken BK48x9. This chip has many capabilities and it is advised you skim the Datasheet.

The frequency range of the radio in this operating mode is from 18 MHz to 1300 MHz , however there is a "Dead Spot" from 630 MHz to 756 MHz that the chip cannot receive or transmit on.

While Radio mode shares many similarities with other radio models and firmware, providing a familiar environment, there are some notable differences in how nicFW operates that may feel unfamiliar at first.

Positions (VFOs)

The radio can operate on up to three frequencies or channels simultaneously. Historically, these ?positions? are called VFOs (Variable Frequency Oscillators) and are labelled A, B, and C from top to bottom.

alt

Each VFO can operate in its own independent mode.

VFO Modes

- Frequency Mode

In Frequency Mode, any valid frequency can be manually entered and used. This mode provides full flexibility for direct frequency operation.

- Valid frequencies are between 18 MHz to 1300 MHz
- Channel Mode

The radio can store up to 999 pre-programmed channels. Channel Mode provides access to these stored channels for quick selection and operation. See: Main Menu -> Channels

- Group Mode

The pre-programmed channels can also be organized into groups (sometimes referred to as ?zones? in other radios). Group Mode allows browsing and operating channels according to these groupings for easier management and navigation. See: Radio -> Channel Groups

Channel Names

When operating in Channel Mode or Group Mode, the channel?s configured name is displayed on the VFO. Each channel supports a name of up to 30 characters, however the amount of text that can be shown on screen depends on the selected font and UI layout.

If a channel name exceeds the available display space, the text may be scrolled horizontally to allow the full name to be read. Scrolling behavior is configurable and can be adjusted or disabled entirely.

See:

- Main Menu -> Advanced -> Name Scroll
- RMS -> UI

Signal Meter

The signal meter displays the strength of incoming signals. It consists of a graphical bar and a numeric S-unit readout, as well as more precise dBm values for both the signal level and the signal floor.

A thin auxiliary bar indicates the amount of Ex-Noise (external noise) currently being received. The meter also displays the current radio state (RX or TX), and during transmission the signal meter doubles as a power output meter.

The signal meter is fully customizable by the user. By default, it appears as shown below:

alt

See:

- RMS -> UI
- Main Menu -> Display -> SBar Style

Battery Level

The radio screen displays the current battery level. The style of the battery read out is configurable by Main Menu -> Display -> Batt Style.

(RT-880 Only) A Lightning Bolt icon will be shown next to the battery readout when the battery is being charged.

Controls

SP-RED (RT-880) | SP-VM (RT-950)

Switches the active position (VFO) between Frequency, Channel, and Group modes.

- To switch to Channel Mode, at least one pre-programmed channel must exist.
- To switch to Group Mode, at least one pre-programmed channel must be assigned to a group.

SP-# (RT-880) | SP-ABC (RT-950)

Cycles the active position (VFO) between A, B, and C.

Note: RT-880 only. nicFW has historically used the above key combinations consistently across all supported radios.

However, the RT-880's native firmware implements these functions in reverse. To accommodate this difference, a setting is provided to swap the behavior of SP-RED and SP-#. See: Main Menu

-> Advanced -> #/Red Mode

RT-880:SP-GREEN / RT-950-Pro:SP-BLUE

Opens the Main Menu.

RT-880:LP-GREEN / RT-950-Pro:LP-BLUE

Opens the Function Menu.

UP / DOWN

Behaviour depends on the active mode:

- Frequency Mode : Increases or decreases the active position's frequency by the configured step size.

See: Main Menu -> VFO -> Step

- Channel Mode : Selects the next or previous pre-programmed channel.
- Group Mode : Selects the next or previous pre-programmed channel that is a member of the currently selected group.

LEFT / RIGHT (RT-950 Only)

Behaviour depends on the active mode:

- Frequency Mode : Increases or decreases the active position's frequency by 10 times the configured step size.

See: Main Menu -> VFO -> Step

- Channel Mode : Skips 10 channels back or forward.
- Group Mode : Selects the next or previous group.

PTT (or Muti-PTT)

If transmission is permitted, begins transmitting on the selected frequency or channel (on the relevant VFO in the case of Multi-PTT). Also see: Main Menu -> VFO -> Multi-PTT

Numeric Keys (0-9)

(Note: 0 is not used for channel selection in Group Mode on the RT-880)

- Frequency Mode

Begins entry of a new frequency. The first key pressed becomes the first digit; subsequent keys enter additional digits. Press to enter a decimal point Press GREEN or # to confirm the entry Press RED to abort entry

- Channel Mode

Begins entry of a channel number from 1 to 999. Press GREEN or # to confirm the selection. Press RED to abort entry. If the entered channel does not exist, the closest configured channel is selected.

- Group Mode

Functions the same as Channel Mode. If the entered channel is not a member of the selected group, the closest channel within that group will be selected instead.

0 and (RT-880 Group Mode Only)

Selects the previous or next group.

Transmit Controls

While PTT is engaged, additional keys become available to perform transmit-specific functions.

DTMF Transmission

When transmitting, the keypad can be used to send DTMF tones.

Key mappings:

- 0-9, , # -> Send their respective DTMF tones
- GREEN -> A
- UP -> B
- DOWN -> C
- RED -> D

Side Key S1 - Manual APRS Beacon

When APRS is enabled, pressing S1 during transmission sends a manual APRS beacon on the currently transmitting frequency.

Warning: Important

This does not transmit on the VFO assigned for APRS operation.

The beacon is sent strictly on the active transmit frequency.

See: Radio -> APRS

Side Key S2 - Repeater Tone

Pressing S2 transmits a legacy repeater access tone.

- Default tone: 1750 Hz
- The tone frequency is user-configurable

Warning: Note

If you are transmitting using S1 or S2 with Multi-PTT, this key gets redefined to the main PTT key.

See: Main Menu -> Advanced -> Repeater Tone

Key Functions and the Function Menu

Key Functions are accessed via LP-GREEN, which opens a dedicated menu containing radio-mode-specific functions. Each function in this menu may also be assigned directly to a key press and customized by the user. For more information on key customization, see: RMS -> User Keys and Main Menu -> User Keys When a function is selected in the menu, its currently assigned key (if any) is displayed beneath the menu for reference.

A video guide regarding the Function Menu can be viewed [HERE](#)

Edit TX Frequency

Default Key: LP-RED Allows entry of a split (offset) transmit frequency. Frequency entry follows the same rules as RX frequency entry. Enter an absolute TX frequency by keying in the value directly. Enter an offset TX frequency by pressing UP or DOWN to indicate the entered value is relative to the RX frequency. When an offset is selected, a + or - symbol appears next to the input field. Press to enter a decimal point Press GREEN or # to confirm entry Press RED to abort. Press S2 to cancel split frequency mode and remove the TX offset.

A video guide demonstrating how to do this can be viewed [HERE](#)

Reverse TX / RX

Default Key: LP-5 When operating with a split or offset TX frequency, this function swaps the RX and TX frequencies. When active, an R indicator appears in the VFO display. Activating the function again restores the original configuration.

Busy Lock

Default Key: LP-6 Prevents transmission when the squelch is open. When active, a padlock symbol is displayed on the VFO.

Squelch Override

Default Key: SP-S1 Forces the squelch open regardless of received signals or tones. While active, the RX/TX indicator area displays OV. Activate the function again to cancel the override.

Multiwatch

Default Key: SP-9

Toggles between Off, On and Group settings for Multi Watch. When enabled, the radio continuously monitors all three VFOs. Upon receiving a signal, the radio switches to the active VFO. After the signal ends, the radio either:

- Remains on the detected VFO, or
- Returns to the previously active VFO
- This behaviour is controlled by Menu -> VFO -> MW Keep VFO.

If the VFO being checked is in Group Mode and Multiwatch is set to Group then all channels in

the VFO's selected group will also be watched.

- This function is best used with smaller groups. Since each channel in the group must be scanned individually, larger groups will increase the scan cycle time and may significantly impact signal detection latency.

Multiwatch only monitors VFOs that are of the same HF or non-HF band. I.e. if the active VFO is set to a frequency over 70 MHz then only VFOs that are also over 70 MHz will be monitored. Similarly if the active VFO is less than 70 MHz only other VFOs that are also less than 70 MHz will be monitored.

A circular arrow icon is shown when Multiwatch is active. If this setting is set to Group then the icon will have a letter G inside it.

Key Lock

Default Key: LP- Locks and unlocks the keypad. When locked, a key icon is displayed on the screen and only PTT and the key assigned to Key Lock remain functional. Note: If Menu -> Advanced -> PIN is enabled:

- PTT is disabled
- The configured PIN must be entered to unlock the keypad

Edit RX Frequency

Default Key: LP-0 In Channel and Group modes, numeric keys are assigned to other functions and cannot initiate RX frequency editing directly. This function must be used to begin editing the RX frequency of a channel.

Enter Group

Default Key: LP-# Available in Group Mode. Allows direct entry of a group number:

- 1 = Group A
- 2 = Group B
- 3 = Group C etc..

Scan (Menu)

Default Key: LP-3 Starts a frequency (VFO) or channel scan.

- Frequency Mode: Scans frequencies using configured scan settings in the Main Menu.

See: Menu -> Scanning and Radio -> Scanning -> VFO Scan

- Channel Mode: Scans all channels that do not require switching to or from the HF band.
- Group Mode: Scans all channels in the selected group that do not require HF switching.

(Essentially, if the active VFO is in the HF band, then the scanner will only scan other HF channels, if the active VFO is not in the HF band, then the scanner will only scan other non-HF channels) See : Radio -> Scanning -> Channel/Group Scan

Scan (VFO)

Default Key: LP-4 Available in Frequency Mode only. Starts a scan using the current VFO settings, rather than the scan settings defined in the menu.

Shut Down (RT-880 Only)

Default Key: LP-EMG Places the radio into a safe shutdown state so it may be powered off. This is only required when the red flash deferral icon is displayed. See Radio -> Storage Deferral

XB Repeater

Default Key: LP-S2 Enables Cross-Band Repeater mode. Requirements:

- VFO A and VFO B must be set to different bands (VHF and UHF)
- Both VFOs must be permitted to transmit

Operation:

- Signals received on either VFO are retransmitted on the other
- Both VFOs appear active
- An XB icon is displayed while active

Day / Night Mode

Default Key: SP-EMG Toggles the LCD between Day and Night display modes. Night mode is typically dimmer. Brightness levels for both modes are configurable. See: Menu -> Display -> Day/Night Level A crescent moon icon is displayed in Night mode.

Frequency Counter

Default Key: LP-7 Also known as a Fast Frequency Scanner on other radios. Rapidly scans for strong signals and locks onto them quickly. The scan is limited to the band (HF or non-HF) on which it is initiated. Main Menu -> VFO -> Tone Monitor will be temporarily set to 'Clone' if it not already set.

This function may only be started in Frequency Mode See: Radio -> VFO Modes

A video guide on using this feature can be found [HERE](#)

Invert LCD

Default Key: LP-S1 Toggles the LCD between normal and inverted display modes. Inverted mode can improve readability in bright ambient light. See: Main Menu -> Display -> Inverted

Si4732 Tuner

Default Key: LP-8 Enters Tuner mode. See: Radio Modes -> Tuner

Spectrum Scope

Default Key: LP-2 Enters Spectrum Scope mode. See: Radio Modes -> Scope

Scan Presets

Default Key: LP-1 Displays a menu of pre-configured Frequency Mode scan presets.

- Use UP/DOWN to browse presets
- Press RED to exit without starting a scan

See: RMS -> Scanning and Main Menu -> Scanning

DTMF Presets

Default Key: SP-S2 Displays a menu of pre-configured DTMF tone sequences.

- Use UP/DOWN to browse sequences
- Press RED to exit without transmitting
- LP-GREEN to edit the name of the selected sequence.
- PTT to begin transmitting and send the selected sequence.

See: RMS -> DTMF and Main Menu -> DTMF

Chan 2 VFO

Default Key: None Available in Channel and Group modes only. Copies the currently selected channel into Frequency Mode on the VFO.

DTMF Speed Dial

Default Key: None When assigned to LP-0 through LP-9, transmits the DTMF preset associated with the corresponding numeric slot.

GPS

Default Key: None Enters GPS operating mode. See: Operating Modes -> GPS

Enter DTMF

Default Key: None Allows manual entry of a DTMF sequence for transmission. Key mapping:

- 0-9, , # -> Enter themselves
- GREEN = A
- UP = B
- DOWN = C
- RED = D
- PTT to transmit
- S2 to cancel

Beacons

Default Key: None Displays a list of received APRS beacons.

- Beacons are numbered #1-#50. #1 is the most recently received
- Use UP/DOWN to browse
- Press RED to exit
- Press GREEN to set the beacon location as the current GPS navigation target.

See: Radio -> APRS

Weather

Default Key: None If a Weather APRS packet has been received, displays the decoded weather data. Press RED to exit. See: Radio -> APRS

TX Power Up / TX Power Down

Default Key: None Increases or decreases the transmit power level of the active VFO. See: Main Menu -> Channel -> TX Power

Focus

Default Key: None Toggles Focus Mode on and off. Focus mode changes the layout of the main display so that only the active VFO is visible. The VFO is stretched to fill the whole screen.

See Main Menu -> Display -> VFO Dim Level

FM

Default Key: None Switch the modulation to FM. See: Main Menu -> Channel -> Modulation

AM

Default Key: None Switch the modulation to AM. See: Main Menu -> Channel -> Modulation

DSB

Default Key: None Switch the modulation to DSB. See: Main Menu -> Channel -> Modulation

Modul Toggle

Default Key: None Cycle the modulation between FM -> AM -> DSB in that order. See: Main Menu -> Channel -> Modulation

Freq Mode

Default Key: None Switches the active VFO directly into Frequency Mode.

Channel Mode

Default Key: None Switches the active VFO directly into Channel Mode. Note: At least one pre programmed channel must exist to enter Channel Mode.

Group Mode

Default Key: None Switches the active VFO directly into Group Mode. Note: At least one pre programmed channel must be a member of any group to enter Group Mode.

KISS Mode

Default Key: None Switches the radio into KISS mode. See:

- Operating Modes -> KISS

Scan MGroups (Multiple Group Scan)

Default Key: None Starts a Group Mode scan that can operate across multiple groups simultaneously. Unlike the standard group scan, which scans only a single group, this function allows you to scan up to four groups at once. When started, you will be prompted to enter up to four group letters (A-Z). The scanner will then cycle through each selected group in turn. This feature is useful for monitoring several related channel groups without needing to merge them into a single group.

- Notes
- Channels may belong to more than one group.
- If the selected groups share common channels, those channels will be scanned more frequently, as they appear in multiple group cycles.
- Only valid (used) groups will be accepted.
- Only groups that have at least one channel that matches the current VFO' frequency band, either below or above 70 MHz will be accepted.

See:

- Radio -> Scanning -> Multi Group Scanning
- Radio -> T9 Text Entry

APRS On/Off

Default Key: None Enables and Disables APRS & GPS operation with a single toggling function. When re-enabling, the function will restore the same APRS mode that was active when APRS was previously disabled using this function. See:

- Main Menu -> APRS -> Enabled

Keypad LED

Default Key: None Enables/Disables the keypad backlight. See:

- Main Menu -> Advanced -> Keypad LED

Beacon NOW

Default Key: None Immediately sends an APRS beacon on the assigned APRS VFO or Channel 999.

Wipe Beacons

Default Key: None Erases all previously received APRS beacons from the radio's memory.

Tuner Mode Overview

Tuner Mode is a receive-only operating mode that utilizes the highly flexible Si4732 tuner chip built into the RT-880. This mode supports reception of Broadcast FM, Long Wave (LW), Medium Wave (MW), and Short Wave (SW) radio bands. Some bands require the use of a secondary antenna, as detailed below.

Available Bands

Band	Frequency Range	Modulation	Antenna
Broadcast FM	64 - 108 MHz	Wideband FM	Primary Antenna
Long Wave (LW)	103 - 522 kHz	AM / USB / LSB / CW	Secondary Antenna
Medium Wave (MW)	530 - 1700 kHz	AM / USB / LSB / CW	Secondary Antenna
Short Wave 1	1.700 - 6.000 MHz	AM / USB / LSB / CW	Secondary Antenna
Short Wave 2	6.000 - 12.000 MHz	AM / USB / LSB / CW	Secondary Antenna
Short Wave 3	12.000 - 30.000 MHz	AM / USB / LSB / CW	Secondary Antenna

Regionally, the Long Wave band can have different step points. This can be fixed by defining the start of the band. See: Main Menu -> Advanced -> LW Start

Spectrum Scope

Tuner Mode can optionally display a spectrum scope showing activity around the tuned frequency while simultaneously receiving audio. This feature has several important limitations:

- On the FM Broadcast band, the scope functions correctly using only a single antenna.
- On Short Wave 3, the scope requires dual antennas and only operates correctly on frequencies above 18 MHz.
- The spectrum scope uses the BK4819 radio SoC to scan surrounding frequencies rather than the Si4732 tuner.

Because of this architecture, enabling the scope may introduce audible interference. The scope can be disabled if this interference becomes undesirable (see controls below).

MultiWatch Interaction

If MultiWatch is enabled in Radio Mode, Tuner Mode reception may be interrupted:

- When a signal on the radio side opens squelch, tuner reception is paused.
- Once the signal is lost, operation automatically returns to Tuner Mode.

This functionality is only available when the Spectrum Scope is disabled, as the scope requires exclusive use of the radio-mode chip.

Signal Meter

Tuner Mode uses the same signal strength meter as Radio Mode to display received signal strength. However, due to limitations of the Si4732:

- No noise bar is available
- No dBm readout is provided

Signal meter updates may also introduce audible interference (typically perceived as periodic beeping). The signal meter can be disabled if required (see controls below).

RDS

The Si4732 supports RDS in FM Broadcast mode. Data such as the station name, category and

status text will be displayed on screen when it is received. RDS Data is also used to set the radio's clock.

Channels

The tuner mode supports channels just like the radio mode does. These channels can be programmed via the RMS or directly by the radio by opening the Main Menu See:

- RMS -> Channels
- Main Menu -> CMS
- SP-RED

Controls

Numeric Keys (0-9)

- Frequency Mode: Initiates entry of a new frequency
- Channel Mode: Initiates entry of a new channel number
- In SW modes, entering a frequency outside the current band automatically switches to the appropriate band (if applicable)

Additional entry keys:

- - Enter a decimal point
- GREEN or # - Confirm entry
- RED - Abort entry

UP / DOWN

- Steps the tuned frequency up or down by the current Step value in Frequency Mode
- Stepping beyond the limits of the current band wraps around to the opposite end
- In Channel Mode, steps between programmed Tuner Channels

and # - BFO Adjustment

USB, LSB, and CW modes only.

- Adjusts the BFO fine-tune (clarifier)
- increases the offset
- # decreases the offset
- Short press adjusts by one step
- Long press continuously steps

The current BFO value is displayed at the bottom of the display as BFO: +000.

SP-GREEN

Opens the Main Menu

LP-GREEN

Opens the Tuner Function Menu, which lists all available tuner functions along with their associated key presses.

- Functions may also be executed by selecting them and pressing GREEN
- Press RED to close the function menu

LP-2 - Reset BFO

Resets the BFO fine-tune value back to zero.

SP-RED

Toggles between Frequency Mode and Channel Mode.

- Tuner channels share the same 999 slots as Radio Mode channels
 - Channels are specific to Tuner Mode
 - At least one Tuner Channel must be programmed to switch to Channel Mode
-

LP-3 - Seek Forward

Frequency Mode only.

Steps upward through frequencies until a sufficiently strong signal is detected.

- Works on all bands
 - Most effective on the FM Broadcast band
 - Limited effectiveness on LW, MW, and SW bands
-

LP-1 - Seek Reverse

Same as Seek Forward, but in the opposite direction.

LP-4 - Set Step

Allows entry of a new frequency step value.

The current step size is displayed at the bottom of the display as Step: X kHz.

LP-5 - Signal Meter On / Off

Toggles the signal meter.

- When enabled, periodic meter updates may cause audible interference
 - Interference is more noticeable on lower bands
 - Usually not present on the FM Broadcast band
-

LP-6 - Squelch On / Off

Enables or disables an automatic noise-gating system.

- Cuts static on empty frequencies
 - Displays an S icon when active
 - Works best on the FM Broadcast band
 - Not recommended for sideband or CW reception
-

LP-7 - Spectrum Scope On / Off

Enables or disables the spectrum scope.

- Disabling the scope eliminates scope-related audio interference
 - Also reduces display clutter for users who find it distracting
-

LP-8 - Power Line Filter On / Off

Toggles the Si4732 built-in Power Line Filter, which attempts to reduce interference from mains

power lines.

The current state is displayed as PLF: On or PLF: Off.

LP-9 - Toggle Bandwidth

AM reception only.

Cycles through available AM bandwidths from 1 kHz to 6 kHz.

The current bandwidth is displayed as BW: X kHz.

SP-EMG

Toggles between Day and Night display modes. Duplicate function of Key Functions -> Day / Night Mode.

LP-RED

Exits Tuner Mode and returns to Radio Mode.

LP-0 - Spirit Box

Rapid audio cycling of the entire band.

This is not a scan; the tuner steps through the band as quickly as possible at the current step rate, providing a fast audio sweep. Useful for estimating band activity and locating potentially interesting signals.

This feature also replicates the function of the so called "Spirit Talker" device, a complete load of nonsense, but what the hell.

SP-S1

Cycles between available bands: FM, LW, MW, SW1, SW2, SW3. The band name is displayed above the tuned frequency.

SP-S2

Cycles between demodulation modes: AM, USB, LSB, CW.

- Has no effect on the FM Broadcast band
- The current modulation mode is displayed under the tuned frequency

Spectrum Scope Mode Overview

The Spectrum Scope provides a graphical view of signal activity across a range of frequencies. It can be started by executing Key Functions -> Scope (default key: LP-2) while in frequency mode.

alt

Frequency Entry

0-9 - Enter Centre Frequency

Begins entry of a new centre frequency in MHz.

- enters a decimal point
- GREEN or # confirms entry
- RED cancels entry

General Controls

SP-RED - Exit Scope

Exits Spectrum Scope and returns to Radio Mode.

SP-GREEN - Main Menu

Opens the main menu.

LP-GREEN - Function Menu

Opens the Scope Function Menu. The menu lists all available scope functions and their associated key shortcuts. Functions may also be executed directly from this menu.

Navigation

UP - Roll Left

Increments the centre frequency upward by the current step value.

- Short press: one step
- Long press: continuous stepping

DOWN - Roll Right

Same as Roll Left, but in the opposite direction.

SP-# - Set Strongest

Sets the centre frequency to the currently strongest detected signal.

Monitoring Functions

Certain actions and events will pause the scope display and switch the radio into receive mode on a user request or detected signal. Reception continues until the the user intervenes or the signal is lost in the case of trigger events.

SP-S2 - Monitor High

Opens the squelch and receives on the strongest signal currently visible on the scope.

SP-S1 - Monitor Centre

Opens the squelch and receives on the current centre frequency.

LP-1 to LP-9 - Set Trigger Level

Sets the signal strength threshold required to trigger monitoring.

A horizontal blue line appears on the scope display indicating the trigger level.

- 1 - Lowest trigger threshold
- 9 - Highest trigger threshold

LP-0 - Cancel Trigger

Removes the active monitor trigger. The blue trigger line will disappear.

SP-# - Set Ignore

Adds the currently monitored frequency to the ignore list so it will not trigger again.

LP-RED - Clear Ignores

Clears all entries from the ignore list.

Scope Configuration

LP-# - Bar Count

Toggles the number of bars displayed in the scope, which also controls the visible frequency span.

Values: 16, 24, 48, 60, 80, 120

LP- - Edit Step

Allows entry of a new frequency step value in kHz.

- enters a decimal point
- GREEN or # confirms entry
- RED cancels entry

Modulation Selection

LP-S1 - Set FM

Sets modulation mode to FM.

LP-S2 - Set AM

Sets modulation mode to AM.

LP-EMG - Set DSB

Sets modulation mode to DSB.

VFO Interaction

PTT - Frequency to VFO

Sets the active VFO in Radio Mode to the current center frequency. This action does not exit Spectrum Scope Mode.

GPS Mode Overview

For radios equipped with GPS capability, GPS Mode provides both a diagnostic readout and a basic navigation environment.

For GPS related settings see Main Menu -> GPS

Data Display Readout

The Data Display is shown first when entering GPS Mode. It presents real-time information received from the GPS chip in an easy-to-read format.

alt

Displayed fields:

- Latitude
- Longitude
- Altitude
- Locked Satellites
- Maidenhead Locator
- Velocity
- Course (heading)
- Time

Additionally, the bottom of the display shows:

- Target Latitude
- Target Longitude

Navigation Environment

The Navigation environment is a simple direction-finding system. It displays a compass with a blip representing a target location, if one is selected. Beneath the compass are several informational fields (see image below for reference).

alt

The radio does not contain a magnetic compass. Direction is determined from motion, meaning the compass updates based on the direction of travel.

Two rotation modes are available:

- The compass needle remains fixed while the outer ring rotates
- A circular arrow icon is displayed top left.
- The outer ring remains fixed while the needle rotates
- No icon is displayed top left.

When a target location is set, a blip appears on the compass indicating the target's direction and distance. If the target lies beyond the current compass radius, the blip is drawn on the outer ring to indicate direction only.

Target information is shown at the bottom of the display, including:

- Target name (if set via a waypoint)
- Distance to target
- Heading to target

An satellite dish icon is displayed top right representing the current GPS status.

- Not Shown
- GPS is disabled
- Dish Pointing Down

- GPS is enabled but not locked
- Dish Pointing Up
- GPS is enabled and locked

Rudimentary Walkie-Talkie Operation

GPS Mode allows very limited radio operation. The VFO that was active when GPS Mode was entered remains active for receive and transmit.

Most Radio Mode features are unavailable. GPS Mode is not intended for normal communication; this functionality exists primarily for testing and diagnostic purposes.

Controls

SP-# - Data / Nav

Cycles between the GPS Data Display and the Navigation environment.

SP- - Rotate Mode

Cycles between compass ring rotation mode and compass needle rotation mode.

When ring rotation mode is active, a circular arrow icon appears in the top-left corner of the navigator screen.

SP-7 - Waypoints

Opens the Waypoint browser.

See:

- Radio -> GPS Waypoints

SP-RED - Exit

Exits GPS Mode and returns to Radio Mode.

SP-9 - Units

Cycles the units used for distance, speed, and related measurements.

See:

- Main Menu -> GPS -> Units

LP-4 - Set Target

Sets the current GPS position as the navigation target.

LP-5 - Clear Target

Clears the currently selected navigation target.

LP-1 - Set Range

Allows editing of the compass radius. This defines the real-world distance represented by the compass ring.

SP-6 - Beacons

Duplicate function. Opens the APRS beacon browser. Received beacons may be selected as navigation targets.

See:

- Key Functions -> Beacons

SP-8 - Weather

Duplicate function. Opens the APRS weather information screen.

See:

- Key Functions -> Weather

LP-2 - MHL Target

Opens a text input field to enter a Maidenhead Locator string to be used as a navigation target.

See:

- Maidenhead Locator System
- Radio -> T9 Text Entry

LP-3 - MHL Position

Opens a text input field to enter a Maidenhead Locator string to be used as a spoofed GPS position.

Main Menu -> GPS -> Enabled must be set to Off for this function to work.

The spoofed location will remain set until:

- GPS is re-enabled
- Another location is spoofed
- The radio is powered off.

See: Radio -> T9 Text Entry

LP-0 - Dec / DegMin

Cycles the latitude and longitude display format between:

- Decimal degrees
- Degrees and minutes

LP-GREEN - GPS Function Menu

Opens the GPS Function Menu. This menu lists and executes all available functions and displays a reminder of their activation key presses.

KISS Mode Overview

When KISS Mode is enabled, the radio operates as a TTNC (Transceiver and Terminal Node Controller). This mode requires a 38,400 baud serial connection to a host system using the standard programming cable and software capable of communicating via the KISS protocol.

See: Main Menu -> APRS -> KISS Mode

Operation

When KISS Mode is active, the radio's display switches to a simplified layout showing:

- A text indicator confirming KISS Mode is active
- The current working frequency
- Settings Values
- TX-Delay in milliseconds
- Persist probability (0-255)
- Slot-Time delay in milliseconds
- TX Power level
- Debug Messages

Control

Working Frequency

The working frequency may be changed by keying in a new frequency, using the same method as in VFO Frequency Mode.

TX-Delay, Persist and Slot-Time

Initially these settings inherit radio mode values

- TX-Delay inherits Main Menu -> DTMF -> Start Pause
- Persist inherits Main Menu -> APRS -> Persist
- Slot-Time inherits Main Menu -> APRS -> Slot Time

They can later be adjusted by the HOST using KISS commands.

TX Power

Inherits the power level of the currently active VFO when KISS mode starts. The level can be changed by pressing the UP and DOWN buttons on the keypad.

Backlight

The LCD backlight may be turned off to preserve battery charge by pressing #

Note about Audio

Regardless of the setting configured in Main Menu -> APRS -> Hear Tones, no audible tones are produced while operating in KISS Mode, the radio is fully muted.

Packet Handling

- Incoming AX.25 packets received on the working frequency are decoded by the radio and forwarded to the host system using standard KISS encoding

- The host system may send KISS-encoded AX.25 packets to the radio, which are then transmitted on the working frequency

Exiting KISS Mode

To exit KISS Mode and return to normal radio operation, press the RED key on the radio. The HOST may also terminate KISS mode by sending the "Exit" KISS command.

More Information

For a technical description of the KISS protocol see the PDF link below [Kiss-TNC PDF](#)

Channel Menu Overview

The Channel Menu configures per-position (VFO) operating parameters that affect how the radio receives and transmits on the selected frequency or channel.

Bandwidth

Sets the FM receive bandwidth for the active position (VFO).

Options:

- Wide - 12.5 kHz
- Narrow - 6.25 kHz
- Automatic - Uses the default bandwidth defined by the band plan for the current frequency

See:

- Main Menu -> Band Plan -> Bandwidth
- RMS -> Band Plan

Modulation

Configures the receive modulation mode.

Warning: Transmission Note

The radio is only capable of native FM transmission. When set to AM or USB, the radio attempts to simulate AM transmission by over-modulation and frequency offset.

This may partially work, but results in poor audio quality and significant adjacent-channel interference. AM transmission is not recommended.

Options:

- FM
- AM
- DSB
- Automatic - Uses the default modulation defined by the band plan for the current frequency

See:

- Main Menu -> Band Plan -> Modulation
- RMS -> Band Plan
- Menu -> Advanced -> AM Hacks

TX CTCSS

Sets the CTCSS tone used during transmission (commonly for repeater access).

- Range: 0.1 to 300.0 Hz
- Exact values (including non-standard tones) may be entered manually
- Use to enter a decimal point
- Standard tones can be browsed using UP/DOWN

When enabled, a CT indicator appears on the VFO display.

See:

- Main Menu -> VFO -> Tone Monitor

TX DCS

Sets the DCS code used during transmission.

- Range: D001N to D777I
- Exact values (including non-standard codes) may be entered manually
- Standard codes can be browsed using UP/DOWN
- Press # to toggle between normal and inverted polarity

When enabled, a DT indicator appears on the VFO display.

See:

- Main Menu -> VFO -> Tone Monitor

RX CTCSS

Sets the CTCSS tone required for reception.

- Range: 0.1 to 300.0 Hz
- Exact values (including non-standard tones) may be entered manually
- Standard tones can be browsed using UP/DOWN
- A special value of 105.0 enables NOAA alert monitoring

When set:

- Squelch will only open if the received signal includes the matching tone
- A CR indicator appears on the VFO display

See:

- Radio -> NOAA
- Main Menu -> VFO -> Tone Monitor
- Main Menu -> Calibration -> XTAL 671

RX DCS

Sets the DCS code required for reception.

- Range: D001N to D777I
- Exact values (including non-standard codes) may be entered manually
- Standard codes can be browsed using UP/DOWN
- Press # to toggle between normal and inverted polarity

When set:

- Squelch will only open if the received signal includes the matching code
- A DR indicator appears on the VFO display

See:

- Main Menu -> VFO -> Tone Monitor
- Main Menu -> Calibration -> XTAL 671

Busy Lock

Controls transmit lockout while a signal is present.

- When On, PTT cannot be engaged while a signal is being received
- A received signal may still be present even if squelch is closed due to RX CTCSS/DCS filtering

When enabled, a padlock icon appears on the VFO display.

PTT ID

Controls transmission of a DTMF identification sequence at the beginning and/or end of a

transmission.

The sequence transmitted is defined by DTMF preset slot #99.

Options:

- Off - PTT ID disabled
- BoT - Send sequence at the beginning of transmission
- EoT - Send sequence at the end of transmission
- Both - Send sequence at both beginning and end

When enabled, the standard PTT-ID icon appears on the VFO display.

See:

- RMS -> DTMF
- Radio -> Main Menu -> DTMF

TX Power

Sets the RF output power for the active position (VFO).

Actual output levels depend on the configured power table.

Options:

- N/T - No transmit (PTT disabled)
- ZERO - Transmit enabled with near-zero output (useful for testing)
- VLO - Very low power
- LOW - Low power
- MID - Medium power
- HIGH - High power
- VHI - Very high power

This setting is always displayed on the VFO.

Warning: ? Note

The band plan may override this setting if a frequency range defines a maximum allowed transmit power.

See:

- Radio -> Main Menu -> Band Plan
- RMS -> Power
- RMS -> Band Plan

Reversed

Swaps the RX and TX frequencies.

- When active, an R indicator appears on the VFO display

See: Radio -> Main Menu -> Reverse RX TX

Clarifier

Applies a fine frequency offset (in kHz) without changing the displayed frequency.

Useful for tuning signals that are slightly off-frequency.

- Use to enter a decimal point
- Use # to toggle between positive and negative offset

Scrambler

Applies simple audio scrambling to transmitted and received audio.

When active an S icon will appear on the display.

Warning: Legal Notice

Audio scrambling is illegal on amateur and public-access bands in many regions.

This feature is intended for commercial use only and provides no real privacy, as it is easily decoded.

- Adjustable range: 2600 Hz to 4500 Hz

Channel Name

Channel and Group modes only.

Allows customization of the channel name. Because the maximum channel length of 30 characters cannot fit in the menu in a single line, the text is split into two lines of 15 characters.

See:

- Main Menu -> Advanced -> Name Scroll
- Radio -> T9 Text Entry
- RMS -> UI
- RMS -> Channels

CMS (Channel Management System) Menu

The Channel Management System (CMS) menu provides tools for creating, deleting, naming, and organizing channels and channel groups.

Channel 999

This can be a special channel reserved for APRS use. This is the channel to configure with your APRS operating frequency. See: Main Menu -> APRS -> Enabled

Write To

Saves the currently active VFO or Tuner configuration as a channel.

- The source may be:
 - A frequency-mode VFO configuration
 - An Si4732 Tuner Frequency. See Operating Modes -> Tuner
 - An existing currently selected channel (effectively copying it)
- Select a channel number from 1 to 999 and press GREEN to confirm
- The menu's Extra Info field indicates whether the selected channel slot is in use or free
- If the source was a tuner frequency that supports RDS, the station name will be set as the new channel's name.

Delete

Erases a stored channel and returns it to a free state.

- Operates in the same manner as Write To
- Select the channel number to erase and press GREEN to confirm
- The Extra Info field indicates whether the selected channel is currently in use

Channel Name

Duplicate menu entry.

See: Main Menu -> Channel -> Channel Name

Add Group / Rem Group

Channel and Group modes only

Adds or removes group membership for the currently selected channel.

- Use the UP/DOWN keys to select a group letter (A-Z)
- Press GREEN to confirm the action

A channel may belong to multiple groups (up to 4) simultaneously.

Adding groups to Tuner based channels is pointless.

The extra info field will display the selected channel's current group membership.

See: Radio -> Channel Groups

Group Name

Group mode only

Sets a custom name for the currently selected group.

- Maximum length: 12 characters

- Press GREEN to confirm the new name

Note: Group names may also be configured with the RMS

See:

- Radio -> T9 Text Entry
- Radio -> Channel Groups
- RMS -> Groups

Squelch Overview

Squelch is a system designed to mute the main speaker until a valid signal is received. nicFW's squelch implementation is significantly more advanced than that of most radios and allows for fine-grained control over how and when audio is unmuted.

nicFW squelch operates based on two independent factors:

- Signal strength (RSSI)
- Received noise level (ExNoise)

For squelch to open, the signal strength must rise above a configured threshold and the noise level must fall below its threshold. For squelch to close, the signal must drop back below its threshold and the noise level must rise again.

While the Squelch Menu is open, the menu's Extra Info field displays real-time numerical values for the current channel or frequency's RSSI and ExNoise levels. These live readings can be used to determine optimal values for several squelch parameters and Signal Floor calibration.

Squelch Menu Options

Level

- Available Values: 0 (Off) to 9
- Default Value: 2

Controls the primary signal-strength requirement for opening squelch.

- 0 (Off)

Disables squelch entirely; the speaker remains unmuted at all times.

- 1

Special noise-only mode. This is the most sensitive setting, but also the most prone to false openings and closings.

- 2-9

Correspond to S-levels:

- 2 -> Minimum of S1
- 3 -> Minimum of S2
- 4 -> Minimum of S3
- and so on

Noise Trigger

- Available Values: 0 to 127
- Default Value: 48

Defines the maximum noise level that must be met for squelch to open. Noise must fall below this value before audio is unmuted.

Noise Hysteresis

- Available Values: 0 to 20
- Default Value: 4

Defines a buffer zone between squelch opening and closing based on noise level.

Example:

- Noise Trigger = 48

- Noise Hysteresis = 4

Squelch will:

- Open when noise drops to 44
- Close when noise rises to 52

This prevents rapid toggling in marginal conditions.

RSSI Hysteresis

- Available Values: 0 to 20
- Default Value: 4

Defines a buffer zone between squelch opening and closing based on signal strength.

- Units are in 1/2 dBm
- Example:
 - Level set to 3 (S2)
 - RSSI Hysteresis set to 4

Squelch will:

- Open at S2 + 2 dBm
- Close at S2 - 2 dBm

Throttle

- Available Values: 0 to 9
- Default Value: 2

Sets a minimum time, in tenths of a second, that must pass after a squelch state change before another change is allowed.

This prevents rapid squelch chatter in noisy environments.

Noise Ceiling

- Available Values: 0 to 127
- Default Value: 60

Defines an absolute maximum allowable noise level.

- Any noise at or above this value will immediately close squelch
- Overrides hysteresis and tail settings

This setting is the primary reason for nicFW's exceptionally fast squelch cutoff behavior.

Squelch Tail Elimination

- Available Values: Off, RX, TX, Both
- Default Value: Off

Reduces squelch tail noise (static bursts) at the end of transmissions.

This system works by transmitting a special 55.0 Hz CTCSS tone at the end of a transmission. If the receiving radio detects this tone, it immediately mutes the speaker before the carrier drops.

Options:

- RX - Respond to tail elimination tones
- TX - Transmit tail elimination tones
- Both - Enable both receive and transmit behaviour

This feature requires compatible settings on both stations.

Tail

- Available Values: 0.0 to 5.0 seconds
- Default Value: 0.5

Defines how long the speaker remains unmuted after a signal drops before squelch closes.

Although often overridden by Noise Ceiling, this setting can help prevent squelch chatter when signals hover near the threshold.

Scanning Menu Overview

For a general overview of scanning behaviour and modes, see: Radio -> Scanning

This menu defines the parameters used for Frequency (VFO) Scanning, as well as global scan behaviour.

VFO Start

Sets the start frequency, in MHz, used when a Frequency Mode scan is initiated.

VFO Range

Defines the width of the scan range, in MHz.

The scan end frequency is calculated as: $\text{End Frequency} = \text{Start} + \text{Range}$ The End Frequency is displayed in the 'Extra Info' field of the menu.

VFO Step

Sets the step size, in kHz, between each scanned frequency.

Scan Return

Defines which frequency the VFO will be set to when a scan is stopped.

Available options:

- Last Signal

Sets the VFO to the last detected active signal.

- Last Scanned

Sets the VFO to the last frequency checked by the scanner.

- Start

Returns the VFO to the scan start frequency.

VFO Modulation

Sets the modulation mode used during a Frequency Mode scan.

Available options:

- FM
- AM
- DSB
- Automatic

When set to Automatic, the modulation is selected based on the band plan default for the start frequency.

See:

- Main Menu -> Band Plan -> Modulation
- RMS -> Band Plan

VFO Bandwidth

Sets the receiver bandwidth used during a Frequency Mode scan.

Available options:

- Wide
- Narrow
- Automatic

When set to Automatic, the bandwidth is selected based on the band plan default for the start frequency.

See:

- Main Menu -> Band Plan -> Bandwidth
- RMS -> Band Plan

Save Ignores

Controls whether the scan exclusion list persists.

- On

Excluded frequencies are saved between scans and power cycles.

- Off

The exclusion list is cleared at the start of each new scan.

See:

- Radio -> Scanning -> Exclusions
- RMS -> Scanning -> Exclusions Tab

Clear Ignores

Immediately clears the current scan exclusion list.

See: Radio -> Scanning -> Exclusions

Ultra Scan Level

Controls how aggressively the Ultra Scan acceleration system operates.

- The default value of 12 is suitable for most environments.
- In noisy RF environments, this value may require adjustment.

Tuning Guidance

Perform a frequency scan and observe the green LED on top of the radio:

- Mostly slow flashes -> Value is too low
- Constant fast flashes -> Value is too high
- Mostly fast flashes with occasional slow flashes -> Optimal

See: Radio -> Scanning -> Ultra Scan

Ultra Scan Time

Sets the analysis time, in microseconds, that Ultra Scan spends evaluating each frequency.

- Default: 1500
- This is an advanced setting and should only be adjusted by experienced users.

Notes:

- If set too low, Ultra Scan may never detect active signals.
 - If set too high, Ultra Scan performance will degrade significantly.
 - This setting directly affects how Ultra Scan Level behaves.
-

Smart Scan

Applies to Frequency Mode scans only.

When enabled:

- Active frequencies are remembered
- The scanner revisits these frequencies more frequently

This can significantly increase scan hits over large frequency ranges, but as the list grows, overall scan speed will decrease.

See:

- RMS -> Scanning -> Smart List
- Radio -> Scanning -> Smart Scan

Smart Select

Allows the user to browse the Smart Scan frequency memory list.

Each slot contains a remembered frequency along with its Heat value (activity score).

When browsing: The selected slot's frequency and heat value are shown in the Extra Info field.

If a slot is empty, the display will show "NOT SET".

- Values: 0-49
- Default: N/A

Smart 2 Chan

Creates a new channel using the frequency stored in the Smart Scan slot selected in Smart Select above.

After selecting this option:

1. Browse to the desired channel number (1-999).
2. Press GREEN to confirm and write the channel.
3. Press RED to cancel.

While selecting a channel number, the Extra Info field will indicate whether the chosen channel slot is already in use. If a used channel slot is selected, the existing channel will be overwritten.

- Values: 0 (Cancel), 1-999
- Default: Cancel

Monitor Time

Defines how long, in seconds, the scanner will monitor a detected signal.

- A value of 0 monitors indefinitely
- Monitoring ends when:
 - This time expires (if non zero)
 - The signal drops
 - The user manually resumes scanning with UP or DOWN
 - The user adds the detected frequency to the exclusions by pressing #

Hold Time

Defines how long, in seconds, the scan will pause after a signal is lost before continuing.

This allows the scanner to briefly 'hang around' to catch replies or follow-up transmissions.

Scan TX Too?

Applies to Channel Mode scans.

- Off (default):

Only the receive frequency of a channel is scanned.

- On:

If a channel has a split or offset TX frequency, that frequency will also be scanned.

Save Preset

Saves the current VFO scan settings as a scan preset.

- Up to 99 preset slots are available
- Select a slot and press GREEN to save
- The menu's Extra Info field indicates whether a slot is Free or In Use

See: RMS -> Scanning -> Presets

Load Preset

Loads a saved scan preset into the current VFO scanning settings.

- Select the desired preset slot
- Press GREEN to load

Erase Preset

Deletes a scan preset.

- Select the preset slot to erase
- Press GREEN to confirm deletion

Stop Any Key

Enables additional keys to stop a scan in progress. When On, any key not assigned to a function while scanning will act like RED and stop the scan.

VFO Menu Overview

The VFO Menu contains settings that control frequency stepping, transmission behavior, audio handling, and multi-VFO operation for the active position.

Step

Sets the frequency step size, in kHz, used when adjusting the VFO frequency with the UP/DOWN keys in Frequency Mode.

This value also determines:

- The spacing between bars in the Spectrum Scope
- The scanning interval used during VFO frequency scans (default scan key: LP-4)

Default value: 12.50 kHz

TX Timeout

Sets the maximum continuous transmission time, in seconds.

- A value of 0 disables the timeout and allows indefinite transmission

Default value: 120 seconds

Mic Gain

Adjusts the sensitivity of the radio's microphone.

- Range: 0 (minimum gain) to 31 (maximum gain)
- This is a relative level with no physical unit

Default value: 25

MultiWatch

Values: Off, On, Group Default value: Off

When enabled, the radio continuously monitors all three VFOs.

Behaviour:

- Upon receiving a signal, the radio switches to the detected VFO
- After the signal ends, the radio either:
 - Remains on the detected VFO, or
 - Returns to the previously active VFO after the time defined by MW Pause
- Note: if any user interaction occurs before this time elapses, the selection will remain on the detected VFO.
- This behaviour is controlled by MW Keep VFO.
- If the VFO being checked is in Group Mode, Multiwatch is set to Group and G-Watch VFOs permits the VFO to operate in Group-Watch mode, then all channels in the VFO's selected group will also be watched.
- This function is best used with smaller groups. Since each channel in the group must be scanned individually, larger groups will increase the scan cycle time and may significantly impact signal detection latency.

Multiwatch only monitors VFOs that are of the same HF or non-HF band. I.e. if the active VFO is set to a frequency over 70 MHz then only VFOs that are also over 70 MHz will be monitored.

Similarly if the active VFO is less than 70 MHz only other VFOs that are also less than 70 MHz will be monitored.

A circular arrow icon is displayed when MultiWatch is active. If this setting is set to Group then the icon will have a letter G inside it.

Multiwatch Suspension

Certain conditions will cause Multiwatch to temporarily stop functioning. When suspended, the icon changes (greyed-out circular arrows with a line through them) to indicate this state. This happens when:

- A user interaction occurs (not PTT) or a signal is received.
- Multiwatch resumes after the period of time defined in MW Pause
- The radio is placed into transmit mode.
- Once transmission is over, Multiwatch resumes when:
 - The user presses a key.
 - After 30 seconds of no signal reception.

G-Watch VFOs (MultiWatch Group Mode VFOs)

Values: A Only, B Only, A And B, C Only, C And A, C And B, All Default: All

Selects which VFOs will perform group watch operations when MultiWatch is set to Group mode.

G-Watch Fast

Values: On, Off Default: On

Controls the scan priority used during Group Watch operation.

When set to On, Group Watch prioritizes scanning the channels within the selected group over the other VFOs and APRS-999. This improves signal detection speed for larger groups but results in slower response times for the single VFOs and APRS-999.

When set to Off, the other single VFOs and APRS-999 are scanned more frequently, keeping them more responsive. The trade-off is that signal detection within the watched group slows down by approximately four times.

MW Pause (MultiWatch Pause)

Defines the delay, in seconds, after a signal loss or user interaction before MultiWatch resumes scanning all VFOs.

Default value: 2.0 seconds

MW Keep VFO

Controls which VFO remains active after a detected signal from MultiWatch is lost.

- On

The detected VFO remains active after the signal ends.

- Off

The radio returns to the previously active VFO after the delay defined by MW Pause.

- Note: if any user interaction occurs before this time elapses, the selection will remain on the detected VFO.

Default value: Off

VOX Level

Enables and configures VOX (Voice-Operated Transmit), allowing transmission without pressing

PTT.

When active a Speech Bubble icon is shown on the display.

- Transmission occurs on the currently active VFO
- Off disables VOX
- Values 1-15 enable VOX
- 1 = least sensitive
- 15 = most sensitive

Default value: Off

VOX Tail

Defines how long, in seconds, a VOX-triggered transmission continues after the user stops speaking.

Default value: 2.0 seconds

Tone Monitor

When enabled, any received signal containing a CTCSS tone or DCS code will display the detected tone/code on the screen. When set on Clone in addition to displaying the detected tone, the active VFO will automatically set its TX Tone to match.

Please use Clone sparingly as it will overwrite your TX subtone settings for whatever channel you're tuned to. You'll be scratching your head as to why all your channels have different CTCSS/DCS settings from what you originally programmed.

Vales: Off, On, Clone Default value: Off

See:

- Main Menu -> Channel -> TX Subtones
- Main Menu -> Advanced -> Tone Monitor Time

Key Frequency

Controls audible keypress feedback.

- Off disables key beeps
- When enabled, the value defines the beep frequency in Hz

Default value: Off

Roger Beep

Controls transmission of a terminating roger beep when a transmission ends.

When active a Music Note icon is shown on the display.

- Off disables the roger beep
- Values from 1 to 9999 define the beep pattern

The pattern consists of a sequence of short tones (musical notes):

- Each digit represents a musical note.
- 0 - B4
- 1 - C5
- 2 - D5
- ..
- 8 - C6
- 9 - D6

- Tones are played in reverse order

Examples:

- 1234 -> F5 E5 D5 C5
- 582 -> D5 C6 G5
- 9 -> D6

Default value: Off

Roger Beep Time

Defines the duration, in milliseconds, of each tone in the roger beep pattern.

Default value: 100 ms

Multi PTT

Configures how PTT buttons map to VFOs.

- Off

Only the main (top) PTT button transmits on the currently active VFO.

- On

All three side buttons function as PTT buttons:

- Top button -> VFO A
- Middle button -> VFO B
- Bottom button -> VFO C

Warning: Note

When Multi PTT is enabled, the short-press and long-press key functions assigned to S1 and S2 are unavailable.

When using S1 or S2 for transmitting, the main PTT button becomes the key to send a repeater tone.

See: Radio Operation -> Repeater Tone

Default value: Off

Calibration Menu Overview

The Calibration Menu provides tools for calibrating battery measurements and correcting oscillator frequency error.

While this menu is open, the Extra Info field displays live diagnostic values to assist with calibration.

Calibration values can be set via the RMS. See RMS -> Calibration

Batt V Cal

Calibrates the displayed battery voltage.

To set this value correctly:

1. Fully charge the battery
2. Adjust this value until the battery voltage shown in the Extra Info field reads 8.4 V

This value is typically around 970, but may vary slightly between radios.

Batt Full

Defines the ADC value corresponding to a fully charged battery.

To calibrate:

1. Fully charge the battery
2. Note the ADC value shown in the Extra Info field
3. Set Batt Full to match that ADC value

This value is typically around 1400, but may vary slightly between radios.

Batt Flat

Defines the ADC value at which the battery is considered discharged.

This is typically the point where:

- The radio still powers on
- Transmission is no longer possible due to insufficient battery voltage

To calibrate:

1. Allow the battery to discharge to this flat state
2. Note the ADC value shown in the Extra Info field
3. Set Batt Flat to match that ADC value

This value is typically around 1000, but may vary significantly between radios.

Batt Low

Defines the ADC threshold at which the battery is considered low.

This level typically corresponds to the point where:

- The radio can still transmit
- However, available battery capacity is nearly exhausted

To calibrate:

1. Allow the battery to discharge until it reaches the desired low state.
2. Note the ADC value shown in the Extra Info field.
3. Set Batt Low to match that ADC value.

This value is typically around 1050, but may vary significantly between radios.

Low Alarm

When enabled, an audible alert will sound every 60 seconds if the battery level is below the threshold defined by Batt Low. If the battery level falls below Batt Flat, the audible alert will sound for twice the normal duration, providing a more urgent warning that the battery is critically low.

- This alarm also sounds if PTT is pressed while the battery is in a low state, however the PTT alarm sounds regardless of if this setting is enabled or not.

XTAL 671

Compensates for manufacturing tolerances in the radio's base crystal oscillator.

The reference oscillator should ideally produce 26.000000 MHz, but in practice it will always be slightly high or low. This error is multiplied as operating frequency increases.

Why this matters

Frequency error scales with frequency multiplication. For example:

- Target frequency: 442 MHz
- Oscillator frequency: 25.99970 MHz
- Multiplier: 17
- Actual tuned frequency: 441.99490 MHz

This results in a 5.1 kHz error.

This is by far the most common cause of 'non-bug' reports where:

- RX CTCSS fails to open squelch
- RX DCS fails entirely

CTCSS generally requires accuracy within ~1 kHz, and DCS within ~500 Hz, making proper calibration essential.

XTAL 671 Quick Calibration Procedure

Requirements

- A trusted reference signal:
- Another radio known to be accurately on frequency, or
- An external signal generator
- A known accurate signal, such as from a repeater.
- The signal must transmit CTCSS or DCS
- Higher frequencies yield better accuracy

Step-by-Step Calibration

1. Ensure Main Menu -> Advanced -> Hunt STones is turned off.
2. Acquire a trusted signal source, this can be from another radio, a signal generator or a broadcast source such as a repeater. This signal source must be sending a valid CTCSS or DCS.
3. Configure the VFO:
 - Set RX CTCSS or RX DCS to match the test signal
 - Set Main Menu -> VFO -> Step to 0.01 kHz
 - Tune to the target frequency

Example: 450.00000 MHz

4. Using UP/DOWN, adjust the frequency until squelch opens reliably.

5. Find the lower limit:

- Step down until squelch no longer opens

Example: 450.00020 MHz

6. Find the upper limit:

- Step up until squelch no longer opens

Example: 450.00050 MHz

7. Calculate the midpoint frequency:

- $(\text{Lower} + \text{Upper}) / 2$

Example: $(450.00020 + 450.00050) / 2 = 450.00035$ MHz

8. Calculate frequency drift:

- $\text{Drift} = (\text{Midpoint} - \text{Target}) \times 100,000$

Example: $(450.00035 - 450.00000) \times 100,000 = 35$

- Note: The resulting value may be negative.

9. Calculate XTAL 671 value:

- $(\text{Drift} \times 671) / \text{Target Frequency}$

Example: $(35 \times 671) / 450 = 52.19 \rightarrow 52$

10. Set XTAL 671 to 52

Notes

- If the calculated value is negative, enter it as a positive number and press # to flip the sign.

Proper XTAL calibration ensures reliable CTCSS/DCS decoding and accurate frequency display across all bands.

Common Whinges

Q: My radio was on-frequency with the factory firmware, why is it off with nicFW?

A: When these radios are manufactured, each one can be tested on the production line and calibration data written to the radio's stock settings. I cannot do that can I? And I have no idea where this data is stored or what format it is in. So calibration has to be performed by the user, I cannot visit each one of you personally to tune your radios or use a time machine, teleport to China and insert myself into the factory to secretly write my own calibration data to new radios.

Common Questions

Q: Why 671? That seems rather arbitrary.

A: It is anything but arbitrary. The value 671 was chosen for a very specific and practical reason.

To accurately correct oscillator drift, the calibration math requires a reference point.

Several factors influenced the choice of this reference:

- Higher reference frequencies provide better adjustment resolution
- Values that are easy for processors to work with are preferred, especially powers of two
- A frequency close to a whole MHz simplifies user-side calculations
- The value must align cleanly with the radio's internal frequency representation

Internally, the radio represents frequencies in 10 Hz units. For example, a frequency of 423.12600 MHz is stored as the integer 42,312,600.

Now consider the frequency 671.08864 MHz.

This value meets all of the design goals:

- It is reasonably high in frequency
- It is very close to a round MHz value (671 MHz)
- Internally it becomes 67,108,864
- 67,108,864 is a power of two

Perfect.

That is why the calibration constant is 671.

Display Menu Overview

The Display menu controls all visual aspects of the LCD, including brightness behavior, battery indicators, signal bars, and power-saving features. These settings allow the display to be optimized for different lighting conditions, usage patterns, and personal preferences.

Batt Style

Sets the visual appearance of the on-screen battery indicator.

Option	Description
Off	Battery readout is hidden
Percent	Displays remaining charge as a percentage
Icon	Displays a graphical bar-style icon
Voltage	Displays battery voltage (up to 8.4 V)

Default: Icon

Timeout

Defines the period of inactivity (in seconds) before the LCD backlight turns off or dims, depending on the Dimmed Level setting. The keypad backlight also turns off when timed out.

- Values: 0 (Disabled) to 999 seconds
- Default: Disabled

When the LCD is turned off, battery saving features are enabled if configured:

- Main Menu -> Advanced -> Battery Save

The Keypad LED will also light in sync with the LCD backlight. This can be disabled however:

- Main Menu -> Advanced -> Keypad LED

Heartbeat

When the LCD is turned off, this setting controls whether the radio periodically flashes an LED to indicate that it is still powered on.

- Values: 0 (Off) to 500 seconds
- Default: 10 seconds

Heartbeat Style

Selects which LEDs are used for the heartbeat indication. If set to flash the RX/TX LED, the following behaviour applies

Battery Level	LED Colour
Over 80%	GREEN
Below 80%, Above Batt Low	YELLOW
Below Batt Low	RED

- Values: Keypad LED, RX/TX LEDs, Both
- Default: Keypad LED

Inverted

When enabled, the LCD colors are inverted. This can improve visibility in bright ambient light conditions.

- Default: Off

See Also: Key Functions -> Invert LCD

Gamma

Adjusts the gamma correction of the LCD (similar to contrast).

- Values: 0 (Least) to 3 (Most)
- Default: 0

SBar Style

Defines the visual style of the signal strength bar.

Option	Description
Stepped	Segmented bar, stepped per S-unit
Solid Step	Solid bar with visible S-unit steps
Solid	Plain solid bar (recommended for inverted mode)
Segmented	Straight segmented bar

Default: Stepped

Day Level

Sets the LCD brightness level when display Mode is set to Day. This should generally be a bright setting.

- Values: 1 (Dimmest) to 31 (Brightest)
- Default: 31

See Also: Key Functions -> Day/Night Mode

Night Level

Sets the LCD brightness level when display Mode is set to Night. This should generally be a dimmer setting.

- Values: 1 (Dimmest) to 31 (Brightest)
- Default: 5

See Also: Key Functions -> Day/Night Mode

Dimmed Level

Defines the LCD brightness level used after the inactivity Timeout expires.

- Values:
 - 0: Disabled (screen turns completely off)
 - 1-31: Brightness level
- Recommended: 1
- Default: Disabled

Mode

Selects the current LCD brightness mode.

- Values: Day, Night
- Default: Day

See Also: Key Functions -> Day/Night Mode

VFO Dim Level

Controls how much inactive VFOs are dimmed relative to the active VFO.

- Values:
- 0: No dimming
- 1-9: Increasing dim level
- 10: Focus Mode

Focus Mode: When set to 10, inactive VFOs are not drawn at all and the active VFO expands to fill the display.

See Also: Key Functions -> Focus Mode

Wake LCD On

Defines which events will wake the LCD when it has timed out.

Option	Description
TX	Wake on transmit or keypad interaction
RX	Wake on received signal or keypad interaction
Keys Only	Wake only on keypad interaction
All	Wake on all events

Default: All

S-Meter Perm

- Values: On, Off
- Default: Off

When set to On, the signal meter is displayed permanently, rather than only appearing when the squelch is open.

Because the dBm readout shares the same screen position as the clock, enabling this option disables the clock display on the main radio screen. The clock is not lost, it will still be shown whenever the Main Menu is opened providing the clock display is enabled.

See: Main Menu -> Time -> Enabled

User Keys Menu Overview

The User Keys menu allows the user to assign custom functions to keypad and side-button presses. Both short and long presses can be configured, enabling quick access to commonly used features.

User key assignments can also be configured using the RMS application. See: RMS -> User Keys

Key

Selects the physical key and press type to be configured by the Function menu below.

- The selected key is shown as a numeric value from 0 to 21 in the standard menu value field
- The Extra Info field displays:
 - The friendly name of the selected key
 - The function currently assigned to it

Key Friendly Names

Each key entry is prefixed with a press type:

- SP - Short Press (pressed and released quickly)
- LP - Long Press (pressed and held for approximately one second)

The press type is followed by the key identifier:

- Grn - Green key (top-left of the keypad)
- Red - Red key (top-right of the keypad)
- 0-9, , # - Standard keypad keys
- S1 - Side button under main PTT
- S2 - RT-880:Bottom side button / RT-950:Second from bottom
- S3 - RT-950 Only:Bottom side button
- EMG - RT-880 Only:Top round orange button

Function

Assigns a function to the key selected in the Key menu.

For a complete list and detailed descriptions of available functions, see: Operating Modes -> Radio -> Key Functions

Save

Unlike most menu settings, user key assignments are not saved automatically.

After completing any changes, select Save to commit the new key mappings to storage. If this step is skipped, all changes will be lost when powering off the radio.

Advanced Menu Overview

As the name implies, these are advanced settings intended for power users. It is strongly recommended that you become familiar with nicFW before making changes in this menu. Here be dragons.

Repeater Tone

While transmitting, pressing side button S2 will send a repeater access tone. This setting defines the frequency of that tone in Hz.

- Default: 1750 Hz
- See: Operating Modes -> Radio -> Transmit Controls

TX Deviation

Sets the FM deviation width using the native units of the BK4819 radio SoC. Refer to the BK4819 datasheet for exact bandwidth conversion.

Most manufacturers using the BK4819 set this value to 64, which is also the default. Beta testing suggests 70 is the upper practical limit before over-deviation becomes problematic.

Increasing this value effectively increases global TX audio gain, but it affects all transmitted audio, including tones, beeps, DTMF, CTCSS, and DCS.

- Range: 0-99
- Default: 64

Subtone Deviation

Controls how loud the transmitted CTCSS and DCS tones are. Increasing this value can help open repeaters in noisy environments.

Units are specific to the BK4819; refer to the datasheet for dB equivalents.

- Range: 0-127
- Default: 74

SPI Deferral (RT-880 Only)

Defines how long configuration and state changes remain cached before being written to persistent storage.

- Range: 0-250 seconds
- Default: 10 seconds
- See: Radio -> Storage Deferral

Tone Monitor Time

Sets how long the detected tone readout remains visible after the signal or tone is lost.

- Range: 2-99 seconds
- Default: 7 seconds
- See: Main Menu -> VFO -> Tone Monitor

AM AGC Fix (RT-880 Only)

A common issue with the BK4819 occurs when receiving strong AM signals: the built-in AGC cannot

attenuate the signal sufficiently, resulting in overdrive.

When enabled, this setting disables the chip's AGC in AM mode and replaces it with a firmware-level RF gain control system. Strong signals may briefly begin with loud buzzing before attenuation takes effect.

- Default: On

AM Emulation Notes

The radio chips used in the RT-880 & RT-950 are the Beken BK48x9. These chips are only capable of FM transmission.

(RT-880)

The RT-880 attempts a crude emulation of AM transmission by combining frequency offset with heavily over-modulated FM. This produces a large deviation in the carrier frequency as the signal is modulated. As the carrier swings above and below the AM receiver's tuned frequency, the signal moves into and out of the receiver's passband. This causes the apparent received signal strength to vary. The AM receiver interprets these variations in signal strength as changes in amplitude, effectively recovering the original modulation from the FM signal.

While intelligible to a receiving station, audio quality is poor and adjacent-channel interference is likely. This feature is not recommended for practical operation.

(RT-950 Pro)

The RT-950-Pro does this differently. It doesn't use the FM adjacent trick that the 880 does. It modulates a carrier by adjusting the output power (and thus the amplitude). Meaning the 950 produces true AM rather than the hack the 880 uses.

AM Hack Transpose (RT-880 Only)

Defines the frequency offset used for AM emulation.

- Units: 10 Hz
- Use # to flip polarity
- Range: -20,000 to +20,000
- Default: +150

AM Hack Displacement (RT-880 Only)

Defines the FM deviation used for AM emulation. Uses the same units as TX Deviation.

- Range: 0-127
- Default: 99

AM TX Range (RT-950 Pro Only)

Defines the vertical normalization window around the TX audio waveform used to control the modulation. In effect, it functions like an inverted microphone gain control: a smaller value produces a tighter window and therefore higher modulation, making the audio louder.

If the value is set too low for the TX audio level, the waveform will exceed the window and be clipped resulting in distortion.

- Units: 12 bit ADC units
 - Range: 100 to 3,000
 - Default: 256
-

Noise Gate

When enabled, this system mutes audio whenever noise reaches or exceeds the value set in Main Menu -> Squelch -> Noise Ceiling.

Using this in combination with fully open squelch provides the most sensitive static gating possible.

- Values: On / Off
- Default: Off

AF Filters

Selects which built-in hardware audio filters are applied to speaker output.

Option	Description
None	No filters applied
De-emf Only	De-emphasis filter only
Lo Only	Low-pass filter only
Lo + De-emf	Low-pass + de-emphasis
Hi Only	High-pass filter only
Hi + De-emf	High-pass + de-emphasis
Hi + Lo	High-pass + low-pass
All	All filters enabled
FSK	Optimized for data decoding

- Default: All

PIN

Sets a four-digit PIN required to unlock the radio.

- Range: 0001-9999
- Default: 0000 (Disabled)
- See: Operating Modes -> Radio -> Keylock

PIN Action

Defines when the PIN is required.

Option	Description
Unlock	PIN required only when unlocking the keypad
Startup	PIN required at radio power-on

- Default: Unlock

Auto Lock

Automatically locks the keypad after a period of inactivity.

- Range: 0-999 seconds
 - 0 = Disabled
 - Default: 0
-

DAC Gain

Individual radios can exhibit different audio amplifier gain levels due to normal manufacturing tolerances. This setting adjusts the output level of the radio's DAC (Digital-to-Analog Converter), effectively changing the usable range of the volume control to make the volume range more consistent.

Increasing the value raises the maximum audio output, which can be helpful in noisy environments such as vehicles where the radio may be too quiet even at full volume. Lowering the value reduces overall output, providing finer control over volume levels in quieter environments.

- Values:
- 0-15 - Increasing output level
- Default: 8

#/Red Mode (RT-880 Only)

Swaps the functions of SP-# and SP-RED.

- nicFW Normal behaviour:
- SP-# cycles active VFOs
- SP-RED switches between Frequency, Channel, and Group modes
- Stock Inverted behaviour; these functions are reversed.
- Default: nicFW
- See: Operating Modes -> Radio -> Controls

LW Start

Defines the starting frequency of the LW (Longwave) band when operating in Tuner Mode. This setting allows the LW band stepping to be adjusted to match different regional and country-specific standards.

- Range: 103 kHz - 211 kHz
- Default: 103 kHz
- See: Operating Modes -> Tuner
- Note: The radio must be power-cycled for changes to this setting to take effect.

Name Scroll

Defines how the UI behaves when a channel name is too long to fit entirely on the display.

- Values
- 0 (Disabled)

No scrolling occurs. Channel names that exceed the available space are simply truncated.

- 1-20

Enables scrolling when the channel name exceeds the available display space by at least the specified number of characters. If the name is shorter than this threshold, scrolling does not occur, and the name will just be truncated (if needed).

- Examples
- A value of 1 causes scrolling for any channel name that does not fully fit on the display.
- A value of 3 causes scrolling only when the channel name is three or more characters longer than the available space.

See

- RMS -> UI
- Main Menu -> Channel -> Channel Name

T9 Advance

Defines the delay, in seconds, after the last key press during T9 text entry before the cursor automatically advances to the next character position.

When set to Disabled, the cursor will not advance automatically and must be moved manually by the user.

- Values: 0.0 seconds (Disabled) to 2.0 seconds
- Recommended: 0.75 seconds
- Default: Disabled

See:

- Radio -> T9 Text Entry

Battery Save

To conserve battery power, the radio can periodically place the RF chip into a low-power state and put the main processor to sleep. While asleep, the radio temporarily reduces activity, then wakes up at regular intervals to check squelch status, perform MultiWatch cycles, and respond to user input.

The duration of this low-power sleep interval is controlled by this setting. Longer sleep periods result in improved battery life, but may make the radio feel less responsive when reacting to incoming signals or user interaction.

Values

- 0.00 - Disabled
- 0.01 to 2.00 seconds - Increasing power-save aggressiveness
- Default: Disabled

Activation Conditions

Battery Save only becomes active when all of the following conditions are met:

- The radio is operating in normal Radio Mode
- Squelch is closed
- The LCD has timed out and is turned off
- The radio is not transmitting

When any of these conditions change, the radio immediately exits low-power mode and resumes full operation.

See

- Main Menu -> Display -> Timeout

Keypad LED

Controls whether the keypad backlight LED is enabled.

By default, the keypad illuminates in sync with the LCD backlight. When this setting is changed to Off, the keypad LED will remain off at all times, regardless of the LCD state.

This can help reduce power consumption slightly or prevent unwanted light in low-visibility environments.

- Values: On, Off
- Default: On

See also:

- Main Menu -> Display -> Timeout

Hunt STones

Enables a more aggressive AFC (Automatic Frequency Control) system to help recover and correctly decode incoming CTCSS tones and DCS codes when the radio or transmitting station is too far off frequency for decoding to occur normally.

This enhanced AFC operates at the firmware level and can compensate for noticeable frequency error, improving tone decoding reliability, particularly on radios that have not been fully calibrated or when receiving slightly off-frequency transmitters.

Because this AFC system actively adjusts reception to ?hunt? for the correct center frequency, it may very briefly make the radio feel slightly less responsive while it locks in. In practice, this effect is rarely noticeable and typically only occurs momentarily while the frequency correction is being established.

- Values: On, Off
- Default: On

Baud Delay

Controls an acknowledgement delay that radio introduces when changing BAUD rates. If higher speeds are not working correctly in the RMS, try increasing this value.

- Values: 100 to 2000 milliseconds
- Default: 100

DTMF Menu Overview

The DTMF menu controls generation, timing, storage, and decoding of Dual-Tone Multi-Frequency (DTMF) signaling. These settings affect manually sent tones, preset sequences, and automatic PTT-ID operation.

Functions specific to DTMF operation are recommended for review:

- Main Menu -> Channel -> PTT ID
- Key Functions -> DTMF Presets
- Key Functions -> DTMF SpeedDial
- Key Functions -> Enter DTMF
- Transmit Controls -> DTMF

DTMF Sequences may also be configured from the RMS

- See: RMS -> DTMF

Deviation

Sets the transmit level (loudness) of generated DTMF tones. Units are specific to the BK4819 radio SoC; refer to the datasheet for conversion to dB.

- Range: 0 to 127
- Default: 112

Digit Time

Defines the duration of each individual DTMF digit in a sequence.

- Range: 0 to 999 ms
- Default: 100 ms

Gap Time

Defines the silent gap between consecutive DTMF digits in a sequence.

- Range: 0 to 999 ms
- Default: 25 ms

Start Pause

Defines the delay between PTT activation and the start of a DTMF sequence.

- Range: 0 to 3000 ms
- Default: 1000 ms

Preset

nicFW provides 99 preset slots for storing pre-programmed DTMF sequences. This menu allows each slot to be populated with a custom sequence.

The Extra Info field indicates whether the selected slot is currently assigned.

Note: The key mappings used to enter DTMF sequences differ from normal keypad input.

Entry Keys:

- 0-9, , # - Enter themselves
- GREEN - A
- UP - B
- DOWN - C

- RED - D
- PTT - Confirm entry
- S2 - Abort entry

Special Slot:

- Slot 99 is reserved for configuration of the PTT-ID sequence.

DTMF presets may also be configured using the RMS application.

- See:
- RMS -> DTMF
- Key Functions -> DTMF Presets

Decoding

Controls the behaviour of the DTMF decoding system. When enabled, detected DTMF tones in received signals are decoded and displayed on screen.

Mode	Description
Off	DTMF decoding disabled
On	Standard DTMF decoding
P-ID	Enables decoding and displays the name of a matching Preset slot when a received sequence matches a stored preset

- Default: Off
- See: Main Menu -> Channel -> PTT ID

Display For

Defines how long decoded DTMF sequences remain visible on the display after the last tone is detected.

- Range: 1 to 99 seconds
- Default: 7 seconds

SeqEnd Pause

Defines the minimum gap between digits that the radio interprets as the end of a DTMF sequence. This is used by the PTT-ID matching system described above.

- Range: 0.1 to 9.9 seconds
- Default: 1.0 second

Time Menu Overview

The radio includes a clock capable of displaying the time of day and, optionally, the day of the week on the screen. The radio does not retain time information when powered off. As a result, if the clock is required, it must be set each time the radio is powered on.

The clock and received signal dBm indicator share the same screen position. Under normal conditions (squelch closed), the time is displayed. When a signal is received and the squelch opens, the display temporarily switches to show the signal strength in dBm.

The clock can be set in the following ways:

- GPS

If GPS is enabled, a successful GPS lock will automatically set the clock. See Radio > Operating Modes -> GPS

- FM Broadcast (RDS)

When in Tuner mode, tuning to an FM broadcast station that supports RDS will automatically set the clock. See Operating Modes -> Tuner

- Manual Configuration

The clock can be set manually using the menu options described below.

Time Zone

When set automatically via GPS, the clock is synchronized to GMT (UTC). This setting applies a positive or negative offset to convert GMT to local time.

- Range: ?12 to +14 hours

Day of Week

Manually sets the current day of the week.

- Available values: MON through SUN

Hours

Manually sets the current hour of the day.

- Range: 0 to 23

Minutes

Manually sets the current minute.

- Range: 0 to 59

Format

Selects how the time is displayed on the screen.

- 24 Hour

Displays the weekday and time in 24-hour format. Example: MON 23:21

- 12 Hour

Displays the time in 12-hour format with an AM/PM suffix. Example: 4:20 PM

Enabled

Turns the clock display on or off. Note: When Main Menu -> Display -> S-Meter Perm is On, the

Time Menu

clock is not shown on the main radio display, instead it is shown under the Main Menu when it is open.

- When Enabled is Off, no time or weekday information is shown on the display or under the Main Menu.

GPS Menu Overview

The GPS menu controls aspects of the radio's Global Positioning System functionality, including waypoint management, navigation targets, unit preferences, and hardware behaviour.

For a more detailed explanation of the radio's GPS capabilities, see Operating Modes -> GPS.

Point Save

Saves the current navigation Target location as a GPS waypoint.

Select one of the 99 available waypoint slots to save to. The Extra Info field indicates whether the selected slot is already in use or free.

- See:
- Operating Modes -> GPS -> Waypoints
- Operating Modes -> GPS -> Navigation
- RMS -> GPS/APRS -> Waypoints

Point Load

Loads a previously saved GPS waypoint and sets it as the current navigation target.

Select one of the 99 waypoint slots to load from. The Extra Info field indicates whether the selected slot contains a waypoint.

- See:
- Operating Modes -> GPS -> Waypoints
- Operating Modes -> GPS -> Navigation
- RMS -> GPS/APRS -> Waypoints

Point Erase

Deletes a stored GPS waypoint.

Select one of the 99 waypoint slots to erase. The Extra Info field indicates whether the selected slot is currently in use.

- See:
- Operating Modes -> GPS -> Waypoints
- RMS -> GPS/APRS -> Waypoints

Enabled

Enables or disables GPS functionality.

When set to Off, the radio will not use the GPS hardware and will instead rely on spoofed location data. This setting defaults to On and must be manually set to Off on non-GPS radio variants.

When enabled, a GPS icon (satellite dish) is displayed on the screen:

- The dish points downward when no GPS lock has been acquired or when lock has been lost.
- The dish points upward when a valid GPS satellite lock is established.

While enabled, the radio will continuously attempt to acquire and maintain a GPS lock whenever the GPS hardware is powered.

- Values: On, Off
- Default: On
- See:
- Radio -> GPS Waypoints

- Operating Modes -> GPS

Units

Selects the preferred measurement system for distance, speed, altitude, and temperature displays.

- Values: Imperial, Metric, Nautical
- Default: Imperial

Reduce QRM

Some radios experience audio interference caused by digital communication between the radio and the GPS chipset. This interference may be present during reception, transmission, or both.

This setting attempts to reduce QRM by temporarily disabling the GPS chip during affected operations.

- Values: None, RX, TX, Both
- Default: None

APRS Overview

Automatic Packet Reporting System (APRS) allows the radio to send and receive digital position and status information.

nicFW on the RT-880 can encode and decode APRS frames even on non-GPS models, which is functionality not offered by even the radio's manufacturer. When used with a GPS-capable radio, the device can function as a full APRS station.

Non-GPS radios can still transmit and receive APRS packets, but positions must be manually configured ("spoofed"). See Operating Modes -> GPS

Received APRS beacons are saved in a list, this list can be browsed using Key Functions -> Beacons

Received weather data is also saved, it can be viewed using Key Functions -> Weather

Waypoints

See Radio -> GPS Waypoints for a full description of GPS waypoints. Waypoint 99 is a special slot, it defines the default spoofed location loaded on startup when GPS is disabled or when using a non-GPS radio. Any selected waypoint can be used as a spoofed location by selecting Waypoint Browser Functions -> Spoof GPS in the Waypoint browser. For configuration using the RMS see RMS -> GPS/APRS -> Waypoints

APRS Menu Overview

The APRS menu controls transmission, reception, and decoding behaviour for APRS packets.

Deviation

Sets the volume (loudness) of transmitted APRS FSK tones. This overrides Main Menu -> Advanced -> TX Deviation.

- Range: 0 to 126
- Default: 64

Beacon Time

Interval between automatic APRS beacon transmissions of the current location.

- Range: 0 (Disabled) to 9999 seconds
- Default: Disabled
- See: Operating Modes -> GPS

Beacon Distance

Minimum distance travelled when GPS is enabled, before a new APRS beacon is sent.

- Range: 0 (Disabled) to 9999 meters
- Default: Disabled
- See: Operating Modes -> GPS

Beacon TX

Sends an APRS beacon at the start or end of a normal PTT transmission, or both. This operates on the current transmit VFO, not the dedicated APRS VFO.

- Values: Off, BoT (Beginning of Transmission), EoT (End of Transmission), Both
 - Default: Off
-

Beacon Comment

Custom comment string transmitted with APRS beacons. Up to 24 characters.

- See: Radio -> T9 Text Entry

Beacon RX Override

When On and APRS Enabled is set to a VFO or 999 , scheduled APRS beacons can interrupt an active reception to transmit.

- Default: Off

Callsign

Set your amateur radio callsign for APRS transmission.

- See: Radio -> T9 Text Entry

SSID

AX.25 secondary station identifier for APRS.

- Range: 0-15
- Default: 7 (Human)

Value	Identifier
0	Primary
1	Vehicle
2	Mobile
3	Camper
4	Home
5	Portable
6	Special
7	Human
8	BBS
9	I-Gate
10	Weather
11	Satellite
12	Digipeater
13	Experiment
14	Truck
15	Emergency

Enabled

Controls if APRS is enabled or not and how APRS operation behaves.

Value	Description
Off	APRS disabled
Active VFO	APRS uses the current active VFO (not recommended for use, mainly for testing, prevents side buttons working)
VFO-A	APRS operates exclusively on VFO-A
VFO-B	APRS operates exclusively on VFO-B

VFO-C	APRS operates exclusively on VFO-C
	Multiwatch needs to be enabled for the exclusive VFO modes
999	APRS operates transparently on channel 999
	Multiwatch is NOT required for 999 mode

Except for Beacon TX and TX Controls -> Manual APRS Beacon.

Operating APRS on a dedicated VFO or 999 is recommended.

APRS Mode	Icon Behaviour
Current VFO & No Automatic Beacons	Compass Icon appears in main icon area. No VFO icon
Current VFO & Automatic Beacons	Compass Icon with central dot appears in main icon area. No VFO icon
Specific VFO & No Automatic Beacons	Compass Icon appears in main icon area and on the selected VFO
Specific VFO & Automatic Beacons	Compass Icon with central dot appears in main icon area and on the selected VFO (without the dot)
Specific VFO	If Hear Tones is set to Off a Speaker Mute icon will also appear in the selected VFO
999 & No Automatic Beacons	999 Icon appears in main icon area. No VFO icon
999 & Automatic Beacons	999 Icon with outward arrows appears in main icon area. No VFO icon

- Default: Off

Hear Tones

Controls whether transmitted and received APRS FSK tones are audible.

Value	Description
On	Tones audible; APRS VFO not muted
Off	Tones muted; APRS VFO muted with mute icon

Decode

Controls whether received APRS packets are decoded.

Value	Description
Off	No decoding
On	Decodes packets and stores in beacon list
Popup	Same as On, but displays a popup on receipt

- See: Key Functions -> APRS Beacons

- Default: Off

Popup Time

Duration for which a popup is displayed when Decode is set to Popup. Popups can be dismissed by the user before the time limit by pressing any key.

- Range: 0.0 (indefinite) to 99.9 seconds

- Default: 3.0 seconds

Symbol

ASCII symbol used in APRS beacons to define station icon. Only standard/primary symbols are supported.

- See: APRS Symbols
- Default: [(? - Person running)

Status

Sets station status encoded in APRS beacons.

Value	Status Message
0	Emergency
1	Priority
2	Special
3	Committed
4	Returning
5	In Service
6	En Route
7	Off Duty

- Range: 0-7
- Default: 7 (Off Duty)

Digipeaters

Configures AX.25 digipeater paths.

Value	Description
None	No digipeaters
WIDE1-1	Generic WIDE1-1 path
WIDE2-2	Uses WIDE1-1 and WIDE2-2
Custom	Custom callsign and SSID via Custom Digipeater and AX25 DP SSID

Custom Digipeater

Callsign used when Digipeaters = Custom.

- See: Radio -> T9 Text Entry

Default: AR0ISS (The international space station)

AX25 DP SSID

SSID used for the custom digipeater.

- Range: 0-15
- Default: 0

Ambiguity

Defines location precision in transmitted APRS beacons.

Value	Description
20 m	High precision, best available
200 m	Medium precision

2 km

Low precision

- Default: 20 m

Filters

Overrides Main Menu -> Advanced -> AF Filters for APRS VFO and 999 reception. Recommended: Low Pass Only. Note: A VFO must be assigned to APRS or APRS 999 mode be in use for this to function

- Default: Low Pass Only

Noise Filter

Extra low-pass filtering for received APRS signals.

- Range: None, 1-3 (increasingly aggressive)
- Default: 2

Noise Reject

Software-level phase hysteresis for APRS decoding.

- Range: None, 1 (less), 2 (more)
- Default: 1

Ignore Own

Prevents the APRS decoder from recording and displaying your own transmitted packets when they are repeated back by digipeaters.

When APRS is in use, your transmitted beacon may be repeated by nearby digipeater stations and received again by your radio. With this setting enabled, those repeated copies of your own packets will be ignored and not added to the received beacon list. This helps keep the beacon log clean and focused on other stations.

- Values: On, Off
- Default: Off

KISS Mode

Enables KISS mode (TTNC) over the serial interface. See:

- Operating Modes -> KISS

Persist & Slot Time

KISS-mode settings:

- Persist: Randomized transmission probability, 1-255
- Default: 128
- Slot Time: Delay in hundredths of a second before retry if Persist fails
- Default: 10 centiseconds

Band Plan Overview

nicFW provides 20 user-definable band slots that can be used to define frequency ranges with associated default behavior. Each band slot can specify default modulation and bandwidth settings, which are used by other parts of the firmware when those settings are configured for Automatic operation. Band slots may also enforce maximum transmit power limits or disable transmit entirely.

When a new frequency is tuned, the band plan slots are evaluated sequentially from slot 0 through slot 19. The first slot where the tuned frequency lies between the configured Start (inclusive) and End (non-inclusive) frequencies is selected and applied.

If the tuned frequency does not match any configured band slot, a default fallback configuration is used:

- FM modulation
- Wide bandwidth
- No power limit
- No frequency wrapping

Default Band Slots

On a newly flashed or factory-reset radio, two band slots are preconfigured:

- Slot 0: AM Aircraft Band (108-136 MHz)
- Slot 1: FM Broadcast Band (88-108 MHz)

Transmit is disabled on both of these bands.

Band Plan Menu

The Band Plan menu allows configuration of individual band slots directly from the radio. While this menu is open, the Extra Info field displays information related to the currently selected slot number.

Band slots may also be configured using the RMS application. See: RMS -> Band Plan

Plan Number

Selects the band slot that will be modified by the other settings in this menu.

- Range: 0 to 19

Start

Sets the starting frequency of the selected band. This value is inclusive.

- Range: 18 to 1300 MHz

End

Sets the ending frequency of the selected band. This value is not inclusive.

- Range: 18 to 1300 MHz

Max Power

Defines a maximum transmit power limit for the selected band. Setting the value to N/T disables transmit entirely for that band.

- Values: N/T, ZERO, VLOW, LOW, MID, HIGH, VHI
- See: Main Menu -> Channel -> TX Power

Modulation

Sets the default modulation for the selected band. This value is used when other modulation settings are configured as Automatic.

- Values: FM AM DSB
- See:
 - Main Menu -> Channel -> Modulation
 - Main Menu -> Scanning -> VFO Modulation

Bandwidth

Sets the default bandwidth for the selected band. This value is used when other bandwidth settings are configured as Automatic.

- Values: Wide Narrow
- See:
 - Main Menu -> Channel -> Bandwidth
 - Main Menu -> Scanning -> VFO Bandwidth

Wrap

When set to On, frequency stepping beyond the limits of the band (using the UP/DOWN keys in frequency mode) will wrap around to the opposite end of the band.

When set to Off, stepping past the band limits will continue into adjacent bands.

Name

Allows a custom name of up to 17 characters to be assigned to the selected band.

- See: Radio -> T9 Text Entry

Save

Changes made within the Band Plan menu are not automatically saved. Selecting this option writes all changes to flash storage.

- Skipping this step will cause all changes to be lost on power cycle
- Note: The radio must be power-cycled after saving for changes to take full effect

Storage Deferral Overview

Note: Only the RT-880 uses storage deferral. The RT-950 Pro does not need it as it has a software power on/off system. Thus this section is only relevant for the RT-880.

Like a PC, smartphone, or tablet, the radio contains non-volatile storage used to preserve its settings and operational state. Users generally expect the radio to return to the same state it was in when powered off, which means that nearly every user action must be saved, channel changes, frequency adjustments, and configuration updates.

However, the storage device used in the radio has a limited write endurance, typically rated for approximately 100,000 write cycles. Exceeding this limit can result in permanent storage failure, rendering the radio unusable.

To protect the storage device and extend the radio's usable lifespan, nicFW employs an aggressive strategy to minimize write operations. This mechanism is referred to as Storage Deferral.

How It Works

When a change would normally require a write to storage, the update is first placed into a temporary buffer rather than being written immediately. Multiple changes can then accumulate in this buffer and are later committed to storage in a single write operation.

By consolidating many small writes into fewer large ones, Storage Deferral dramatically reduces the total number of write cycles over the lifetime of the device.

The amount of time changes remain buffered before being written is user-configurable. See: Main Menu -> Advanced -> SPI Deferral

Impact on User Experience

When there are pending changes waiting to be written to storage, a red warning triangle is displayed on the main screen. If the radio is powered off while this indicator is visible:

- The pending changes will be discarded
- No corruption or damage will occur
- The radio will simply revert to the last saved state on the next power-up

To ensure all changes are preserved, you may:

- Wait for the warning triangle to disappear, indicating that storage has been updated, or
- Use Main Menu -> Shut Down to place the radio into a safe state before powering it off

Channel Groups Overview

Each of the 999 pre-programmed channels can be organized into groups (sometimes referred to as zones) to simplify navigation and management of channels with similar characteristics.

There are 26 available groups, labelled A through Z. Letters are used instead of numbers to avoid confusion with channel numbering. Each channel may be a member of up to four groups.

Adding or Removing Group Membership

On the Radio:

- Enter Channel Mode
- Navigate to the channel you wish to modify
- Open:
 - Main Menu -> CMS -> Add Group - to add the channel to a group, or
 - Main Menu -> CMS -> Rem Group - to remove the channel from a group
- Select a group letter (A-Z) and press GREEN to confirm (or RED to cancel)

The Extra Info field in the menu displays the current group membership of the active channel.

Using RMS: To edit channel group membership using the Radio Management Software, see: RMS -> Channels

Renaming Groups

Each group can be assigned a custom name of up to 12 characters.

On the Radio

- Enter Group Mode
- Select the group you wish to rename
- Open Main Menu -> CMS -> Group Name
- Press GREEN to begin editing the group name
- Press GREEN to confirm once editing is complete
- Or press RED to cancel

Text entry follows the standard T9 input method. See: Radio -> T9 Text Entry

Using RMS To rename groups using RMS, see: RMS -> Groups

Using Groups

Once channels have been organized into groups, they can be accessed using Group Mode.

To enter Group Mode:

- Press RED to cycle the active VFO's operating mode

(or #, depending on Main Menu -> Advanced -> #/Red Mode)

- You are in Group Mode when:

The channel number is displayed in yellow

The selected group's name is visible in the VFO display

Group Mode Behaviour

Group Mode operates similarly to Channel Mode, with the following differences:

- Only channels that are members of the selected group are available
- The # and 0 keys cycle the selected group up or down
- Starting a scan (default: LP-3) scans only channels within the selected group
- The selected group can also be changed using LP-# (default) and entering a number:

1 = A

Channel Groups

2 = B

3 = C etc..

Scanning Overview

The radio supports multiple scanning modes depending on the active operating mode. Scanning can be performed in Frequency (VFO) Mode, Channel Mode, or Group Mode, each with specialized behaviours.

While scanning, the radio steps through frequencies/channels until it detects a signal that is strong enough to break the current squelch settings. At which point the scan stops and the radio begins monitoring the detected signal. See: Main Menu -> Squelch

A scan icon (a radio antenna) is shown on the display during scanning.

During signal scanning, the display DOES NOT update for every scanned channel or frequency. This is because updating the screen takes time and would slow the scan down significantly. To compromise, the display updates every tenth of a second instead, this provides a fast enough visual reference without slowing down the scanner too much. nicFW scans much faster than this however, so it may appear as though channels and frequencies are being missed, this is NOT the case.

There are various menu settings that configure the performance and behaviour of scanning. See: Main Menu -> Scanning

VFO/Frequency Scanning

VFO/Frequency Scanning is used when a scan is initiated in Frequency Mode. A continuous range of frequencies is scanned. The frequencies scanned depend on the function used to start the scan:

- See Main Menu -> User Keys

Scan (Menu) - Default key: LP-3

Scans a range of frequencies defined in the Main Menu.

- Main Menu -> Scanning -> VFO Start defines the frequency the scan starts from in MHz.
- Main Menu -> Scanning -> VFO Range defines the range of the scan in MHz. The scan will run up to: VFO Start + VFO Range
- Main Menu -> Scanning -> VFO Step sets in kHz the gap between scanned frequencies.
- Main Menu -> Scanning -> VFO Modul sets the modulation mode of the scan, FM, AM or DSB or Automatic for whatever the band plan defines.
- Main Menu -> Scanning -> VFO Bandw sets the receive bandwidth of the scan, Wide, Narrow or Automatic for whatever the band plan defines.
- Note: See Main Menu -> Band Plan and RMS -> Band Plan for more information on the above automatic settings.

Scan (VFO) - Default key: LP-4

Uses the current VFO's frequency, modulation, bandwidth and step size instead of the menu-defined scan parameters. The scan range is however still defined by Main Menu -> Scanning -> VFO Range.

Scan Presets - Default key: LP-1

Scans a pre-configured frequency range selected from a menu.

- See: Operating Modes -> Radio -> Scan Presets

Ultra Scan

An acceleration system for VFO scanning that can increase scan speed by up to 10%.

- See: Main Menu -> Scanning -> Ultra Scan Level

Smart Scan

Remembers previously active frequencies and revisits them periodically.

Smart scan is not recommended to be used with Monitor Time enabled, as their root purposes conflict. Neither is it ideal for users who regularly skip detected frequencies by pressing UP/DOWN during monitoring. That use-case is incompatible with the fundamental idea of the smart scan system.

- Results in more hits but can slow overall scan speed as memory builds.
- See: Main Menu -> Scanning -> Smart Scan

Why does the frequency mode scan start lower than the start frequency I configured?

When performing a frequency (VFO) mode scan with Ultra Scan enabled, the scanner must first establish a reliable noise floor so it can correctly identify potential signals.

Ultra Scan does this by maintaining a rolling minimum of the noise levels from the last 16 scanned frequencies. This keeps the noise floor local and relevant to the part of the spectrum currently being scanned, which greatly improves detection accuracy and speed.

However, this approach has a small side effect: at the very beginning of a scan, the noise floor has not yet been fully initialized or localized. The first 16 frequencies would otherwise be evaluated using incomplete or inaccurate noise data.

To avoid this, the scanner automatically shifts the effective start frequency downward by 16 steps. This gives Ultra Scan enough data to establish a proper noise baseline before reaching your configured start frequency.

As a result, the scan appears to begin slightly below the start frequency you set, but by the time it reaches your configured range, the noise floor is fully settled and signal detection is accurate.

Channel Scanning

Used when a scan is initiated in Channel Mode (default: LP-3).

- Steps through each pre-programmed channel at approximately 30 channels per second.

Group Scanning

Used when a scan is initiated in Group Mode (default: LP-3).

- Functions identically to Channel Scanning, except only channels in the selected group are scanned.

Multi Group Scanning

Started by the User Key function Scan MGroups.

- The scan toggles between up to four groups entered by the user.
- The and 0 keys used to change groups during a group scan do not function in multi group scan.
- When stopping a scan, the VFO will switch to channel mode. This is because you might not stop the scan on the right group for the channel Scan Return will set.

See:

- Operating Modes -> Radio -> Scan MGroups
- Main Menu -> Scanning -> Scan Return

Scanning Controls

While a scan is running the following keys perform scanning-specific functions.

Control	Function
UP / DOWN	Change scan direction. During monitoring, skips the monitor and continues the scan.
	Note: When skipping, the monitored frequency is temporarily ignored. This prevents the same active frequency being immediately detected again.
#	Place the current or last detected frequency into the ignore (exclusion) list.
RED	Stop the scan immediately.
/ 0	Group scan only, change group during scanning (single group scan only)
Any Other Key	If Main Menu -> Scanning -> Stop Any Key is On, also stops the scan

Scanning Exclusions

- A list of excluded frequencies can be maintained to ignore nuisance signals.
- The list can be set to:
 - Persistent - Saves exclusions between scans
 - Temporary - Cleared each time a scan is started (configured via Main Menu -> Scanning -> Save Ignores)
- Exclusions can also be manually cleared: Main Menu -> Scanning -> Clear Ignores
- The exclusion list may also be managed via the RMS. See: RMS -> Scanning -> Exclusions

Scanning Limitations

- The radio uses a mechanical relay to switch in the HF band (<70 MHz).
- Rapid switching between HF and non-HF channels during a scan can stress the relay.
- To protect the relay, Channel and Group Mode scans will only scan channels that match the band of the VFO when the scan is started.
- When group scanning, you cannot switch to a group that only contains channels that are in the other band.

Frequency Counter / Fast Frequency Scan

A special scanning mode designed to find very strong signals over a wide frequency range very quickly. See: Operating Modes -> Radio -> Frequency Counter

GPS Waypoints Overview

nicFW can store up to 99 pre-programmed locations, referred to as Waypoints. Waypoints may be used to:

- Save locations while navigating
- Set navigation target locations
- Spoof GPS coordinates

Waypoint #99 is reserved as the default spoofed location when the radio is powered on with GPS functionality disabled. See: Main Menu -> GPS -> Enabled

Waypoint Browser

Waypoints can be browsed by executing the Waypoints function while in GPS Mode (default key: SP-7).

Each waypoint entry displays:

- Waypoint number
- Waypoint name
- Location in degrees/minutes
- Location in decimal degrees
- Maidenhead locator

alt

Browser Controls

SP-RED - Exit

Closes the Waypoint Browser and returns to the previous screen.

SP-GREEN - Set to Target

Sets the currently selected waypoint as the active navigation target.

LP-0 - Edit Name

Creates, edits, or changes the name of the selected waypoint.

See:

- Radio -> T9 Text Entry

LP-1 - Edit Latitude

Edits the latitude of the selected waypoint. Input is in decimal degrees format.

Controls during entry:

- - Enter a decimal point
- GREEN or # - Confirm entry
- RED - Cancel entry

LP-2 - Edit Longitude

Edits the longitude of the selected waypoint. Operation is identical to Edit Latitude, but applies to longitude.

LP-4 - Invert Latitude

Flips the latitude sign between positive and negative (North ? South).

LP-5 - Invert Longitude

Flips the longitude sign between positive and negative (East ? West).

LP-RED - Spoof GPS

Sets the selected waypoint as the spoofed GPS location. Main Menu -> GPS -> Enabled must be set to Off for this function to operate.

The spoofed location will remain set until:

- GPS is re-enabled
- Another location is spoofed
- The radio is powered off.

T9 Text Entry Overview

Some radio settings and configurations require text input. Text is entered using a T9-style input method.

When a text field is active, an input box with a cursor is displayed. Use the UP and DOWN arrow keys to move the cursor right and left, respectively. Use the numeric keypad (0-9) to cycle through the characters available for the current cursor position, following standard T9 key assignments. Use and # to enter their respective characters.

For multi-line inputs, the side buttons S1 and S2 can be used to move the cursor up or down a line.

After typing, the cursor can automatically advance to the next position once key presses stop. This behaviour is controlled by the T9 Advance setting:

- If T9 Advance is enabled, the cursor will move forward automatically after the configured delay.
- If T9 Advance is disabled, the cursor will remain in place until moved manually.

See: Main Menu -> Advanced -> T9 Advance

Press GREEN to confirm the entry. Press RED to cancel the entry.

NOAA Tone Detection

A special RX CTCSS tone value of 105.0 can be assigned to a channel to enable NOAA alert-style tone detection.

When this tone is configured:

- The channel's squelch remains closed until a 1050 Hz tone is detected on the received signal
- Once detected, the squelch opens and remains open
- The squelch will close again when:
 - The signal is lost, or
 - The user manually intervenes

When the channel is tuned to a designated NOAA weather frequency, this configuration effectively enables NOAA alert monitoring.

Visual Indicator

When NOAA Tone Detection is active, a raincloud icon is displayed on the VFO.

Detection Latency

The radio requires approximately 250 milliseconds to detect the presence of a tone.

Because NOAA frequencies typically transmit a continuous carrier:

- The radio cannot immediately reject the signal
- It must wait the full detection period to determine whether the tone is present

As a result, when monitoring NOAA frequencies, particularly while Multiwatch is enabled, the radio may feel slightly less responsive.

Configuration

See: Radio -> Main Menu -> RX CTCSS

CAT Commands

Overview

CAT (Computer Aided Transceiver) allows a radio to be remotely controlled by a host machine over a serial connection using structured command/response messages.

This is distinct from the Remote tab of the RMS, which functions as a screen and keypad mirroring system. CAT operates using a command-driven control paradigm instead.

Connection and framing

Commands are sent as plain text at 38,400 baud over the standard programming interface and must be terminated with ONLY a (carriage return, ASCII 13).

Responses and prompt

Each command returns a text response that is one of the following:

- an error
- an acknowledgement
- a reply to a query

Responses are terminated with (ASCII 13, 10) and followed by the command prompt (if E 2 is active):

cmd:

Each command described below documents its expected responses.

Communication check

Sending a single with no command returns only the prompt (providing E 2 is active):

cmd:

Implementations are encouraged to do this to confirm valid communication before sending commands.

Warning: Note: The list below documents all currently available CAT commands.
Additional commands may be introduced in future revisions based on user requests.

RX Frequency

Command Format:

- F [MMMMKKKHHH]

Command Responses:

- F MMMMKKKHHH

To indicate the actual frequency that was tuned or the reply to a query. Sometimes it can be changed by frequency limits.

- E Invalid Frequency

If the frequency was incorrectly formatted or outside the valid range of 18 to 1300 MHz

Sets or queries the currently active VFO's receive frequency in Hz.

Setting the RX Frequency

- If the VFO is in Channel or Group mode, the selected pre-programmed channel's RX frequency will be permanently updated to the specified value.
- The TX frequency automatically follows the RX frequency and preserves any configured offset.

- The frequency must be supplied in the exact format shown.
- Leading and trailing zeros are required.

Example:

F 0435546000

Sets the VFO receive frequency to 435.546 MHz.

Querying the RX Frequency

To query the active VFO receive frequency, send:

F

TX Frequency

Command Format:

- T [MMMMKKKHHH]

Command Responses:

- T MMMMKKKHHH

To indicate the actual frequency that was tuned or the reply to a query. Sometimes it can be changed by frequency limits.

- E Invalid Frequency

If the frequency was incorrectly formatted or outside the valid range of 18 to 1300 MHz

Sets or queries the currently active VFO's transmit frequency in Hz.

Setting the TX Frequency

- If the VFO is in Channel or Group mode, the selected pre-programmed channel's TX frequency will be permanently updated.
- The frequency must be supplied in the exact format shown.
- Leading and trailing zeros are required.

Example:

T 0443123500

Sets the VFO transmit frequency to 443.1235 MHz.

Querying the TX Frequency

To query the active VFO transmit frequency, send:

T

Active VFO

Command Format:

- V [1,2,3 or A,B,C]

Command Responses:

- V X

Where X is the VFO actually switched to or the reply to a query. Sometimes the system may not allow a switch to a particular VFO.

- E Invalid VFO

If the VFO number was incorrectly formatted or outside the valid range of 1 to 3

Sets or queries the currently active VFO.

Setting the Active VFO

The active VFO may be selected using either numeric or letter identifiers:

- 1 or A -> VFO-A
- 2 or B -> VFO-B
- 3 or C -> VFO-C

Examples:

V 2

Switches to VFO-B.

V A

Switches to VFO-A.

Querying the Active VFO

To query the currently active VFO, send:

V

Active VFO Mode

Command Format: M [0,1,2 or V,C,G]

Command Responses:

- M X

Where X is the actual mode switched to (0, 1 or 2) or the response to a query.

- E Invalid Mode

If the mode was incorrectly formatted or outside the valid range of 0 to 2.

Sets or queries the operating mode of the currently active VFO.

Setting the mode

The active VFO's mode may be set using either numeric or letter identifiers:

- 0 or V -> Frequency Mode
- 1 or C -> Channel Mode
- 2 or G -> Group Mode

Examples:

M 2

Sets the active VFO to Group Mode.

M V

Sets the active VFO to Frequency Mode.

Querying the Active VFO's Mode

To query the currently active VFO's operating mode, send:

M

Active VFO Channel Number

Command Format:

- C [001 to 999]

Command Responses:

- C XXX

Where XXX is the actual channel switched to (001 to 999) or the response to a query.

- E Invalid Channel Number

If the channel number was incorrectly formatted or outside the valid range of 001 to 999.

- E Invalid Mode

If the active VFO is in Frequency Mode.

Sets or queries the channel number of the currently active VFO. This command functions only when the VFO is in Channel Mode or Group Mode. If the specified channel number is not programmed, the nearest valid channel will be selected. When in Group Mode, the specified channel must belong to the currently selected group; otherwise, the nearest channel within that group will be selected.

Setting the Channel Number

The active VFO's channel may be set using a three-digit number with leading zeros.

Examples:

C 007

Sets the active VFO to Channel 7.

C 172

Sets the active VFO to Channel 172.

Querying the Active VFO's Channel

To query the currently active VFO's channel number, send:

C

Active VFO Modulation

Command Format:

- Q [0 to 3]

Command Responses:

- Q X

Where X is the actual modulation switched to (0 to 2) or the response to a query.

- E Invalid Modulation Mode

If the input was incorrectly formatted or outside the valid range of 0 to 3.

- E Invalid Modulation

If the input was malformed.

Sets or queries the modulation mode of the currently active VFO.

Setting the modulation

The active VFO's modulation may be set using a single number from 0 to 3.

Value	Modulation
0	FM
1	AM
2	DSB
3	Automatic

- Note: When set to 3 (Automatic) the radio will select the modulation defined in the Band Plan for the RX Frequency.

Queries and replies will report this modulation (0 to 2). See: Main Menu -> Band Plan & RMS -> Band Plan

Examples:

Q 0

Sets the active VFO to FM.

Querying the Active VFO's modulation

To query the currently active VFO's modulation, send:

Q

Active VFO Bandwidth

Command Format:

- B [W, N, A]

Command Responses:

- B X

Where X is the actual bandwidth switched to (W or N) or the response to a query.

- E Invalid Bandwidth Value

If the input was incorrectly formatted or outside the valid values (W, N, A)

- E Invalid Bandwidth Syntax

If the input was malformed.

Sets or queries the bandwidth of the currently active VFO.

Setting the bandwidth

The active VFO's bandwidth may be set using a single letter.

Value	Bandwidth
W	Wide
N	Narrow
A	Automatic

- Note: When set to A (Automatic) the radio will select the bandwidth defined in the Band Plan for the RX Frequency.

Queries and replies will report this bandwidth (W or N). See: Main Menu -> Band Plan & RMS -> Band Plan

Examples:

B W

Sets the active VFO to Wide bandwidth.

Querying the Active VFO's bandwidth

To query the currently active VFO's bandwidth, send:

B

Active VFO TX Power

Command Format:

- Y [0 to 6]

Command Responses:

- Y X

Where X is the actual TX Power switched to (0 to 6) or the response to a query.

- E Invalid TX Power Value

If the input was incorrectly formatted or outside the valid range (0 to 6)

- E Invalid TX Power Syntax

If the input was malformed.

Sets or queries the transmit power of the currently active VFO.

Setting the power

The active VFO's transmit power may be set using a number from 0 to 6.

Value	Power
0	Transmit Disabled
1	Zero Power
2	Very Low
3	Low
4	Medium
5	High
6	Very High

- Note: TX Power can be overridden by the Band Plan, replies and queries will report the overridden value.

See: Main Menu -> Band Plan & RMS -> Band Plan

Examples:

Y 5

Sets the active VFO TX Power to High.

Querying the Active VFO's TX Power

To query the currently active VFO's transmit power, send:

Y

Active VFO Group

Command Format:

- G [A to Z]

Command Responses:

- G X

Where X is the actual group switched to (A to Z) or the response to a query.

- C XXX

Where XXX is the channel within the group that was switched to (001 to 999).

- E Invalid Group

If the group letter was incorrectly formatted or outside the valid range of A to Z.

- E Invalid Mode

If the active VFO is in Frequency or Channel Mode.

Sets or queries the group of the currently active VFO. This command functions only when the VFO is operating in Group Mode. If the specified group letter is not programmed, the nearest valid group will be selected.

Setting the Group Letter

The active VFO's group may be set using a single letter from A to Z.

Example:

G H

Sets the active VFO to Group H.

Querying the Active VFO's Group

To query the currently active VFO's group, send:

G

PTT

Command Format:

- X [1 or 0]

Command Responses:

- X Y

Where Y is the actual PTT state.

- E Invalid PTT State

If a value other than 0 or 1 was supplied

- E Invalid PTT Syntax

If the command was malformed

Triggers (X 1) and Releases (X 0) the main PTT key.

- Important Note:

The radio transmits significant RF energy, this can interfere with the USB Serial cable and cause it to malfunction or fail. It can also destroy nearby electronics. So if you transmit and then everything stops working THIS IS WHY!

Key Press

Command Format:

- P [0-9,G,R,U,D,.,#,S,T,E,X]

Command Responses:

- P Y

Where Y is the actual key state.

- E Invalid Key State

If a value other than those allowed was supplied.

- E Invalid Key Syntax

If the command was malformed.

Triggers and Releases any of the keypad buttons. Short and Long presses are the duty of the implementing system with respect to timing.

Note regarding Multi-PTT:

- Use P S and P T to engage Multi-PTT on VF0-B and VF0-C.
- Use P X to release Multi-PTT on VF0-B and VF0-C.

Value	Key
0 to 9	0 to 9
G	GREEN
R	RED
U	UP
D	DOWN
	STAR/ASTERISK
#	HASH/POUND
S	S1
T	S2
E	EMG
X	RELEASE KEY

Squelch

Command Format:

- S

Command Responses:

- S X

Where X is the current squelch state 0 = Closed 1 = Open

Queries the current state of the radio's squelch.

Signal & ExNoise

Command Format:

- D

Command Responses:

- D XXX YYY

Where XXX is the current signal level and YYY is the external noise level. Values are 3 digits with leading zeros. Units are in the BK4819's register format.

- Divide by 2 and subtract 160 to convert to dBm

Queries the current signal and external noise levels being received by the radio.

Command Replies

Command Format:

- E [0, 1 or 2]

- Default: 2

This command DOES NOT send any replies.

Selects how the radio responds to commands.

Value	Result
2	The radio sends replies, queries and errors to commands and also sends the cmd: prompt
1	The radio sends only replies, queries and errors to commands, but the cmd: prompt is NOT sent
0	The radio will send NOTHING back to the host at all, no errors, no replies, nothing.

Serial Protocol Overview

Communication between HOST and RADIO is performed over a USB serial programming cable specifically designed for the RT-880. The default communication speed is 38,400 baud, but this can be changed dynamically using a command packet (see below).

All communication uses a synchronous packet-based protocol, with the exception of Remote Mode, which uses its own asynchronous protocol. The serial protocol is used to control link speed and to read from or write to the radio's flash storage.

Packet Structure

All packets begin with:

- Signature byte: 0xAA
- Command ID byte

Any following data is command-specific.

- All multi-byte values are little endian

Change Baud Rate

Changes the active serial baud rate.

Host -> Radio

Data Bytes	Element
1	0xAA - Signature
1	0x70 - Command ID (Change Baud)
4	New baud rate (32-bit unsigned)

Radio -> Host (Acknowledgement)

Data Bytes	Element
1	0xAA - Signature
1	0x70 - Command ID

The radio first replies at the original baud rate, then sends the same acknowledgement again 100 ms later at the new baud rate.

Notes

- The radio defaults to 38,400 baud
- This command must be sent at 38,400 baud
- After 5 seconds of no serial activity, the radio automatically falls back to 38,400 baud
- If higher speeds are required, this command must be sent during initialization

Storage Read and Write

These commands allow the host to access the radio's flash storage. Sequential reads and writes are supported. Due to flash constraints, operations are performed in either 128-byte or 4096-byte blocks.

Read Storage (128 Bytes)

Host -> Radio

Data Bytes	Element
1	0xAA - Signature
1	0x30 - Command ID (Read 128 bytes)
4	Storage address (32-bit unsigned)

Radio -> Host

Data Bytes	Element
1	0xAA - Signature
1	0x30 - Command ID
4	Storage address
128	Storage data
1	Additive checksum of storage data

Write Storage (128 Bytes)**Host -> Radio**

Data Bytes	Element
1	0xAA - Signature
1	0x40 - Command ID (Write 128 bytes)
4	Storage address
128	Storage data
1	Additive checksum of storage data

Radio -> Host

Data Bytes	Element
1	0xAA - Signature
1	0x40 - Command ID

Read Storage (4096 Bytes)**Host -> Radio**

Data Bytes	Element
1	0xAA - Signature
1	0x10 - Command ID (Read 4096 bytes)
4	Storage address

Radio -> Host

Data Bytes	Element
1	0xAA - Signature
1	0x10 - Command ID
4	Storage address

4096	Storage data
1	Additive checksum of storage data

Write Storage (4096 Bytes)

Host -> Radio

Data Bytes	Element
1	0xAA - Signature
1	0x20 - Command ID (Write 4096 bytes)
4	Storage address
4096	Storage data
1	Additive checksum of storage data

Radio -> Host

Data Bytes	Element
1	0xAA - Signature
1	0x20 - Command ID

Finalize Write

Commits all previously written data to flash storage. This should be called after you've sent all write packets for the operation.

Host -> Radio

Data Bytes	Element
1	0xAA - Signature
1	0x48 - Command ID (Finalize Write)

The radio acknowledges with the same packet.

Reboot Radio

Reboots the radio. Required after writing storage and finalizing so changes take effect. This command is not acknowledged.

Host -> Radio

Data Bytes	Element
1	0xAA - Signature
1	0x50 - Command ID (Reboot)

Send Smart List

Requests the contents of the radio's Smart Scan frequency memory.

Host -> Radio

Data Bytes	Element
------------	---------

1	0xAA - Signature
1	0x55 - Command ID (Send Smart List)

Radio -> Host

Data Bytes	Element
1	0xAA - Signature
1	0x55 - Command ID
4	Frequency (32-bit unsigned, 10 Hz units)
2	Heat / activity score (16-bit unsigned)
-	Frequency + Heat repeated 50 times

Start Remote

Starts a Remote Mode session. A baud-rate change is typically performed before issuing this command. This command is not acknowledged.

See: Technical Data -> Remote Protocol

Host -> Radio

Data Bytes	Element
1	0xAA - Signature
1	0x51 - Command ID (Begin Remote Session)

Remote Protocol Overview

Remote Mode uses a datagram-based packet protocol. Communication is unacknowledged (except for PING), meaning packets are not guaranteed to arrive and are not retried.

The radio's display is 240x320 pixels in a portrait orientation. Implementations should ideally keep the key controls, display mirror and any other connection based controls all on the screen together without having to scroll.

Once Remote Mode is active:

- The radio no longer responds to normal serial commands
- Storage read/write operations are disabled
- Only Remote protocol packets are processed

Session Control

START (HOST -> RADIO)

Begins a remote control session.

Bytes: 0xAA, 0x51

PING (Bidirectional)

To keep the remote session alive, the HOST should send a ping byte once per second.

Bytes: 0xAA

The RADIO replies with the same byte (0xAA). This keeps both directions active and prevents timeouts.

- If the HOST stops sending pings, the RADIO will terminate Remote Mode after a few seconds.
- The HOST should terminate Remote Mode if ping replies from the RADIO stop arriving.

EXIT (HOST -> RADIO)

Terminates Remote Mode and returns the radio to normal operation.

Byte: 0x52

Display Update Packets (RADIO -> HOST)

These packets instruct the HOST to update the remote display.

TEXT Packet

Instructs the HOST to draw a text string.

Field	Description
0x55	Packet signature
0x02	Packet type
1 byte (unsigned)	X coordinate
2 bytes (unsigned, little-endian)	Y coordinate
1 byte (unsigned)	Font number
2 bytes (unsigned, little-endian)	Background color (RGB565)
2 bytes (unsigned, little-endian)	Foreground color (RGB565)
Variable	Null-terminated ASCII text

1 byte (unsigned)	Additive packet checksum
-------------------	--------------------------

Fonts

Different font sizes and symbols must be implemented. All fonts are monospaced. They are numbered as follows:

- 0 : 8x8 ASCII Font
- 1 : 8x16 ASCII Font
- 2 : 16x16 ASCII Font
- 3 : 16x24 ASCII Font
- 4 : 24x24 ASCII Font
- 5 : 24x32 ASCII Font
- 6 : 16x16 Symbol Font
- See table below

Symbol Font

ASCII Code	Symbol
32	Regular Space
33	Padlock
34	PTT-ID Icon
35	VOX Icon
36	Scanning Icon
37	Pause Icon
38	UP Chevron
39	Key Icon
40	Circular Arrow (Multiwatch)
41	UP Arrow
42	DOWN Arrow
43	LEFT Arrow
44	RIGHT Arrow
45	Minus Symbol (-)
46	Plus Symbol (+)
47	Warning Triangle
48	Cross Band Repeater Icon (XB)
49	Crescent Moon
50	Rain Cloud
51	Music Note
52	Charging Icon (Lightning Bolt)
53	Filled Circle
54	GPS Not Locked Icon
55	GPS Locked Icon
56	APRS Icon
57	APRS Icon With Beacon
58	Mute Icon
59	ARPS 999 Mode Icon (No Automatic Beacons)

60	ARPS 999 Mode Icon (With Automatic Beacons)
61	Greyed Out Circular Arrow (Multiwatch Temporarily Disabled)
62	Circular Arrow With G (Groupwatch mode)

RECT Packet

Instructs the HOST to draw a filled rectangle.

Field	Description
0x55	Packet signature
0x01	Packet type
1 byte (unsigned)	X coordinate
2 bytes (unsigned, little-endian)	Y coordinate
1 byte (unsigned)	Rectangle width
2 bytes (unsigned, little-endian)	Rectangle height
2 bytes (unsigned, little-endian)	Rectangle color (RGB565)
1 byte (unsigned)	Additive packet checksum

LED Packet

Updates the graphical representation of the radio's status LED.

Field	Description
0x55	Packet signature
0x03	Packet type
1 byte (unsigned)	LED status
1 byte (unsigned)	Additive packet checksum

- LED Status Values
- 0x00 : LED off (black)
- 0x01 : Red LED
- 0x02 : Green LED
- 0x03 : Green + Red (yellow)

Keypad

Implementations should provide controls to mirror the radio's physical keypad. Ideally these controls should be arranged in a similar way to the radio itself.

Side Buttons	1st Column	2nd Column	3rd Column
PTT	Emerg	Up	
PTT	Green	Down	Red
S1	1	2	3
S1	4	5	6
S2	7	8	9
S2		0	#

Note that PTT, S1 & S2 span two rows.

Communicating key press and release events is detailed below.

Key Input Packets (HOST -> RADIO)

These single-byte packets simulate keypad and button events on the radio. The HOST should send pressed packets when the user presses a control and release packets when the user releases the control. PTT has its own release event packet (0xFE), every other key shares a release event packet (0xFF)

Value	Action
0x00	Keypad 1 pressed
0x01	Keypad 4 pressed
0x02	Keypad 7 pressed
0x03	pressed
0x04	Keypad 2 pressed
0x05	Keypad 5 pressed
0x06	Keypad 8 pressed
0x07	Keypad 0 pressed
0x08	Keypad 3 pressed
0x09	Keypad 6 pressed
0x0A	Keypad 9 pressed
0x0B	# pressed
0x0C	GREEN pressed
0x0D	UP pressed
0x0E	DOWN pressed
0x0F	RED pressed
0x10	Side button S1 pressed
0x11	Side button S2 pressed
0x12	Emergency button pressed
0x13	PTT pressed
0xFE	PTT released
0xFF	Any other key released (except PTT)

Packet Checksums

The last byte of all display packets sent by the radio is a checksum. The checksum is calculated by adding every byte of the packet other than the checksum byte itself. The sum is an 8-bit additive checksum, so it wraps at 0xFF. The host should calculate this checksum and check it matches the checksum byte sent at the end of the packet.

For Example:

Here's the Text Packet that draws the Battery Charging icon.

Field	Description
0x55	Packet signature = 85
0x02	Packet type = 2
0xB7	X coordinate = 183
0x27 0x00	Y coordinate = 39, 0
0x06	Font number = 6

0x00 0x00	Background color black = 0, 0
0x1F 0x00	Foreground color blue = 31, 0
0x34 0x00	Charging icon = 52, 0
0x8E	checksum = 142

If we manually check this, add them all together $0x55 + 0x02 + 0xB7 + 0x27 + 0x06 + 0x1F + 0x34$
 $= 0x18E$ $0x18E \% 0x100 = 0x8E$: CORRECT

Implementations of this protocol should use the above sample packet data to test the checksum functionality is correct.

Storage Flash Memory Map

Section	Start Address	Length (hex)
SPIFONT8X8	0x49000	0x0300
SPIFONT8X16	0x49300	0x0600
SPIFONT16X16	0x49900	0x0C00
SPIFONT16X24	0x4A500	0x1200
SPIFONT24X24	0x4B700	0x1B00
SPIFONT24X32	0x4D200	0x2400
SPIFONTSYMBOLS	0x4F600	0x0C00
SPICHANNELS	0x50200	0xF400
SPISETTINGS	0x5F600	0x0400
SPICALIBRATION	0x5FA00	0x0080
SPIPOWERBANDS	0x5FA80	0x0080
SPIGROUPNAMES	0x5FB00	0x0200
SPILONGPRESSES	0x5FD00	0x0100
SPIBANDPLAN	0x5FE00	0x0300
SPIPALETTE	0x60100	0x0100
SPISKIN	0x60200	0x0400
SPILOGO	0x60600	0x25900
SPIPOWERCAL	0x85F00	0x1000
SPISCANPRESETS	0x86F00	0x1000
SPIDTMFPRESETS	0x87F00	0x1000
SPIWAYPOINTS	0x88F00	0x1000
SPIPERMIGNORE	0x89F00	0x0100
SPIBEACONS	0x8A000	0x1A00

All values are little endian

Current Magic Value Is: 0xABC6

- Note: Magic values are subject to change, but I try not to do this if I can help it.

channelInfo Structure Definition

```
'c
#define CHANNAMEMAXLENGTH 30
typedef struct attribute((packed)) { u32 rxFrequency; // 0 u32 txFrequency; // 4 - 10Hz units
u32 groups; // 8 - each 8 bits represents a group number 1-26 or 0 (slot not filled) u16
rxSubtone; // 12 -- 0.1 Hz units for CTCSS u16 txSubtone; // 14 -- DCS code | 0x8000 ( | 0x4000
as well for inverted codes) s16 clarifier; // 16 u8 modulation; // 18 - 0 FM, 1 AM, 2 DSB, 3
Auto u8 bandwidth; // 19 - 0 Wide, 1 Narrow u8 busyLock; // 20 Bool u8 txPower; // 21 u8
reversed; // 22 Bool u8 pttID; // 23 u16 magic; // 24 - If valid, channel is active, if invalid
channel is inactive. u8 scrambler; // 26 u8 reserved[5]; // 27 char name[CHANNAMEMAXLENGTH]; //
32 } channelInfo; '
```

There are a total of 1002 of these structs stacked in the CHANNEL section of the flash memory.
Slots 0,1 & 2 are the VFOs, slots 3 to 1001 are the 999 channels.

settingsInfo Structure Definition

```
'c typedef struct attribute((packed, aligned(4))) { u16 magic; u16 sigbarStyle;
u16 squelch; u16 squelchNoiseLev;
u16 noiseCeiling; u16 noiseHysteresis;
u16 squelchThrottle; u16 step;
u16 txTimeout; u16 micGain;
u16 multiWatch; u16 multiWatchDelay;
u16 battStyle; u16 lcdTimeout;
u16 heartbeat; u16 lcdInverted;
u16 lcdGamma; u16 scrambleFreqDEFUNCT;
u16 vox; u16 voxTail;
u16 squelchTailElim; u16 txDeviation;
u16 stDeviation; u16 keyLock;
struct { u16 groupChannel[27]; u8 lastUsedGroup; u8 lastGroup; u8 isVfo; u8 padding[3]; }
groupMem[3];
u16 spiDeferralTime; u16 toneMonitor;
u16 toneTime; u8 activeVfo; u8 savedDimLevel;
u32 scanStart; u32 scanRange;
u16 scanStep; u16 ultraScan;
u16 scanModulation; u16 scanBandwidth;
u16 scanMonitorTime; u16 amAgc;
u16 scanReversed; u16 lcdDayLevel;
u16 lcdNightLevel; u16 lcdMode;
u16 dtmfDev; u16 repeaterTone;
u16 scanPersist; u16 noiseGate;
u16 afFilters; u16 lcdDimLevel;
u16 keyFreq; u16 rogerBeep;
```

```

u16 rogerBeepTime; u16 multiWatchSwitch;
u16 timeZone; u16 gpsSync;
u16 gpsunitType; u16 gpsrotate;
u16 gpsrange; u16 swapHashRed;
tunerBandInfo tunerBands[6];
u16 tunerLastChannel; u16 aprsHysteresis;
u8 fmTunerSquelch; u8 fmTunerScope; u8 tunerBand; u8 foobargyy11;
u16 rssiHysteresis; u16 squelchTail;
s16 amHackTransform; u16 amHackDisplacement;
u16 vfoDimLevel; u16 dtmfDigitTime;
u16 dtmfGapTime; u16 dtmfStartPause;
u16 dtmfDecoding; u16 dtmfDisplay4;
u16 dtmfEndSeq; u16 timeFormat;
u16 wakeLcdOn; u16 pinCode;
u16 pinAction; u16 timeEnabled;
u16 fskDeviation; u16 beaconTime;
u16 beaconDistance; u16 beaconTx;
char beaconComment[25]; char aprsCallsign[7];
u16 aprsMonitor; u16 aprsSsid;
u16 aprsDecode; u16 audioBoost;
u16 PERSIST; u16 SLOTTIME;
u16 micEabc; u16 micEdigipeat;
u16 micEstdSymbol; u16 micEambiguity;
u16 popupTime; u16 aprsVFO;
u16 aprsAfFilters; u16 ax25DigipeaterSSID;
u16 aprsLowPass; u16 multiPTT;
u16 autoLock;
char ax25Digipeater[7];
u8 gpsCoordType;
u16 gpsReduceQRM;
u16 heartbeatStyle;
u16 saveIgnores;
u16 scanReturn;
u16 ultraScanDelay;
u16 smartScan;
u16 aprsBeaconRxOverride;
u16 lwStartFreq;
u16 scanAbortAnyKey;
u16 scrollTrigger;
u16 alwaysOnRssi;
u16 last; } settingsInfo;

```

tunerBandInfo struct

```

c typedef struct attribute__((packed, aligned(4))) { u32 freq; u32 step; s16 bfo; u8 mode; struct
{ u8 bw: 7; u8 pflilt: 1; } bwConfig; } tunerBandInfo;

```